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Course: Cybersecurity

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Title: Active Directory for OT Environment

Exercise 1: Active Directory Domain Services (AD DS) Verification

Objective

The objective of this exercise was to verify that the Active Directory Domain Services (AD DS) environment had been successfully configured and that the Domain Controller was operating correctly before performing user and access management tasks.

Verification

The Domain Controller (**DC01**) was verified to be operating within the **ot.local** domain. The Active Directory Users and Computers (ADUC) console was opened to confirm that the domain structure was available and that the server was functioning correctly.

The following configuration was verified:

Configuration	Value
Domain Controller	DC01
Domain Name	ot.local
Static IP Address	192.168.1.200
Active Directory Role	Installed and Operational

Result

The Active Directory environment was successfully verified. The **ot.local** domain was accessible through Active Directory Users and Computers, confirming that the Domain Controller was ready for the creation of Organizational Units, user accounts, security groups, and access control policies in the subsequent exercises.

Conclusion

The verification confirmed that the Active Directory Domain Services environment was functioning correctly and provided the foundation required for the remaining laboratory exercises.

Exercise 2: Organizational Unit (OU), Security Groups, and User Account Creation

Objective

The objective of this exercise was to establish a structured Active Directory environment by creating an Organizational Unit (OU), provisioning user accounts, and implementing Role-Based Access Control (RBAC) through Global Security Groups. Organizing users into departmental groups simplifies administration, improves security, and ensures that permissions can be assigned according to organizational roles rather than individual user accounts.

Lab Execution Summary

After verifying the successful deployment of the Active Directory Domain Services environment, a dedicated Organizational Unit (OU) was created to organize user objects within the **ot.local** domain. Two Global Security Groups representing different departments were then created, followed by the creation of six user accounts. Each user account was assigned to its appropriate security group to implement centralized access management using Role-Based Access Control (RBAC).

Implementation Steps

Step 1: Open Active Directory Users and Computers

Logged into the Domain Controller (**DC01**) using the Domain Administrator account.

Opened **Server Manager**, selected **Tools**, and launched **Active Directory Users and Computers (ADUC)**.

Step 2: Navigate to the Domain

Expanded the **ot.local** domain within ADUC to display the available Organizational Units and default containers.

Verified that the domain was functioning correctly before creating additional directory objects.

Step 3: Create the Organizational Unit

Created a new Organizational Unit named:

OT_Users

The Organizational Unit serves as a dedicated container for storing departmental user accounts and security groups, making administration more organized and scalable.

Step 4: Create Global Security Groups

Within the **OT_Users** Organizational Unit, two Global Security Groups were created:

Security Group	Purpose
Engineering_Team	Stores Engineering department user accounts
IT_Team	Stores IT department user accounts

Both groups were configured as:

- **Group Scope:** Global
- **Group Type:** Security

IT_Team Properties ? ✕

General Members Member Of Managed By

 IT_Team

Group name (pre-Windows 2000):

Description:

E-mail:

Group scope

☐ Domain local

☒ Global

☐ Universal

Group type

☒ Security

☐ Distribution

Notes:

OK Cancel Apply

Engineering_Team Properties ? ✕

General Members Member Of Managed By

 Engineering_Team

Group name (pre-Windows 2000):

Description:

E-mail:

Group scope

☐ Domain local

☒ Global

☐ Universal

Group type

☒ Security

☐ Distribution

Notes:

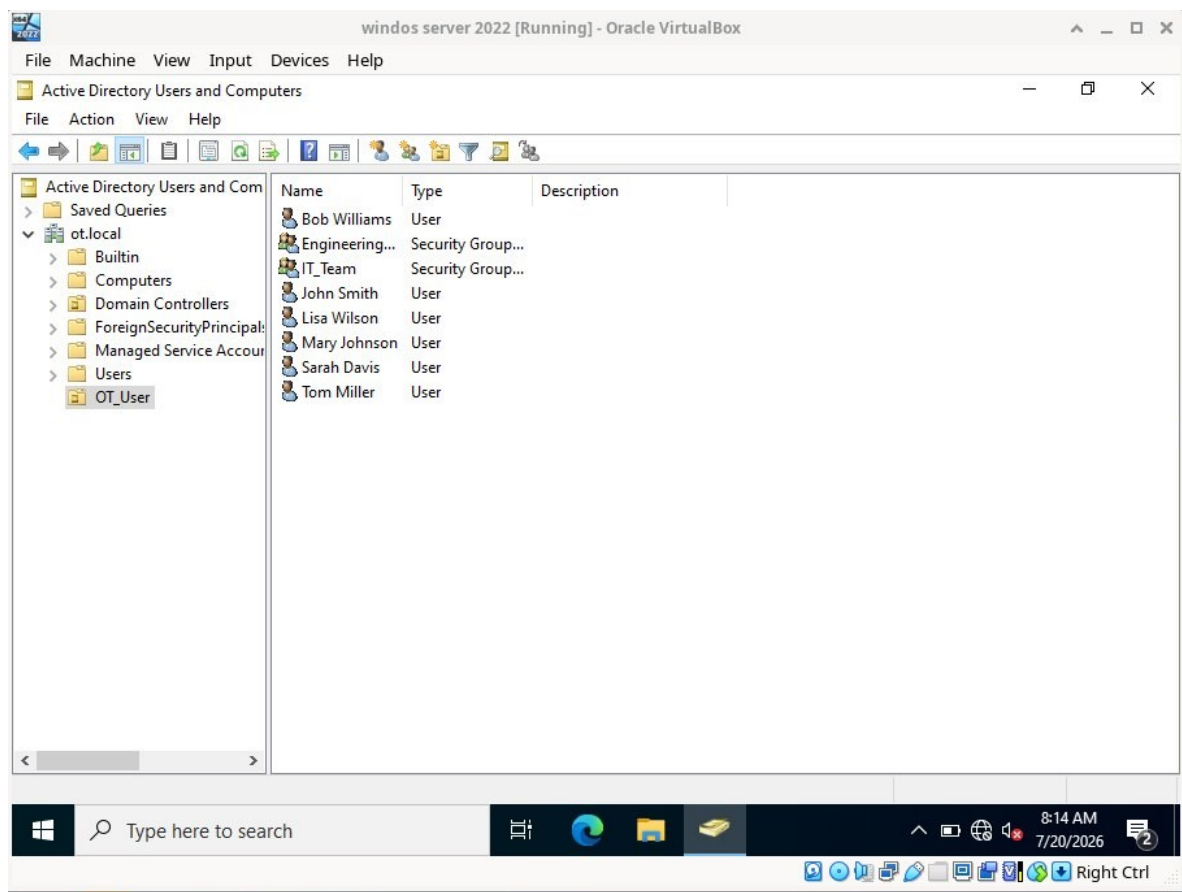
OK Cancel Apply

Step 5: Create User Accounts

Six Active Directory user accounts were created within the **OT_Users** Organizational Unit.

The following user accounts were configured:

Full Name	Username	Department
John Smith	jsmith	Engineering
Mary Johnson	mjohnson	Engineering
Bob Williams	bwilliams	Engineering
Sarah Davis	sdavis	IT
Tom Miller	tmiller	IT
Lisa Wilson	lwilson	IT



Step 6: Assign Users to Security Groups

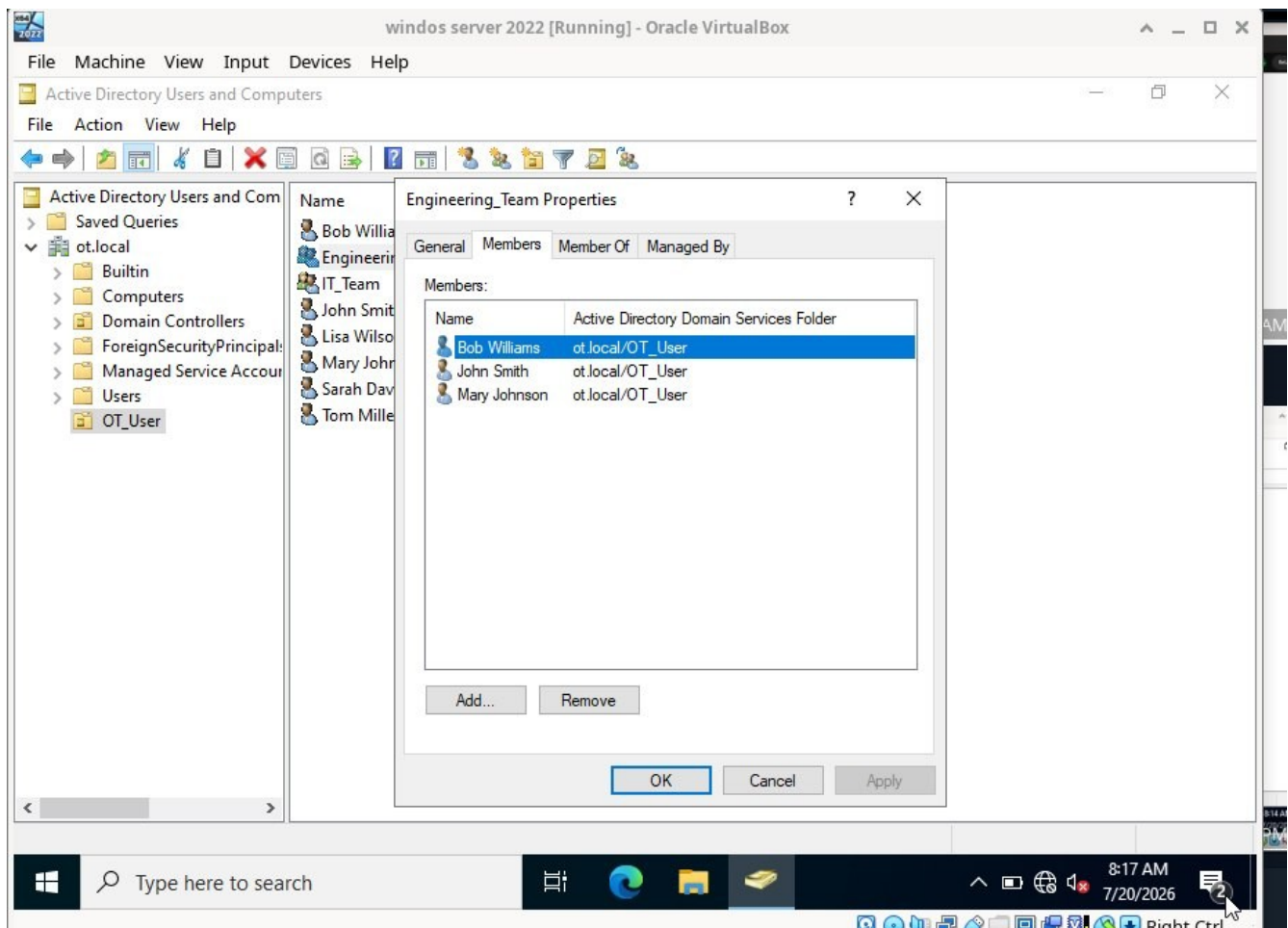
The newly created user accounts were assigned to their respective departmental security groups.

Engineering Team

- jsmith
- mjohnson
- bwilliams

were added to:

Engineering_Team



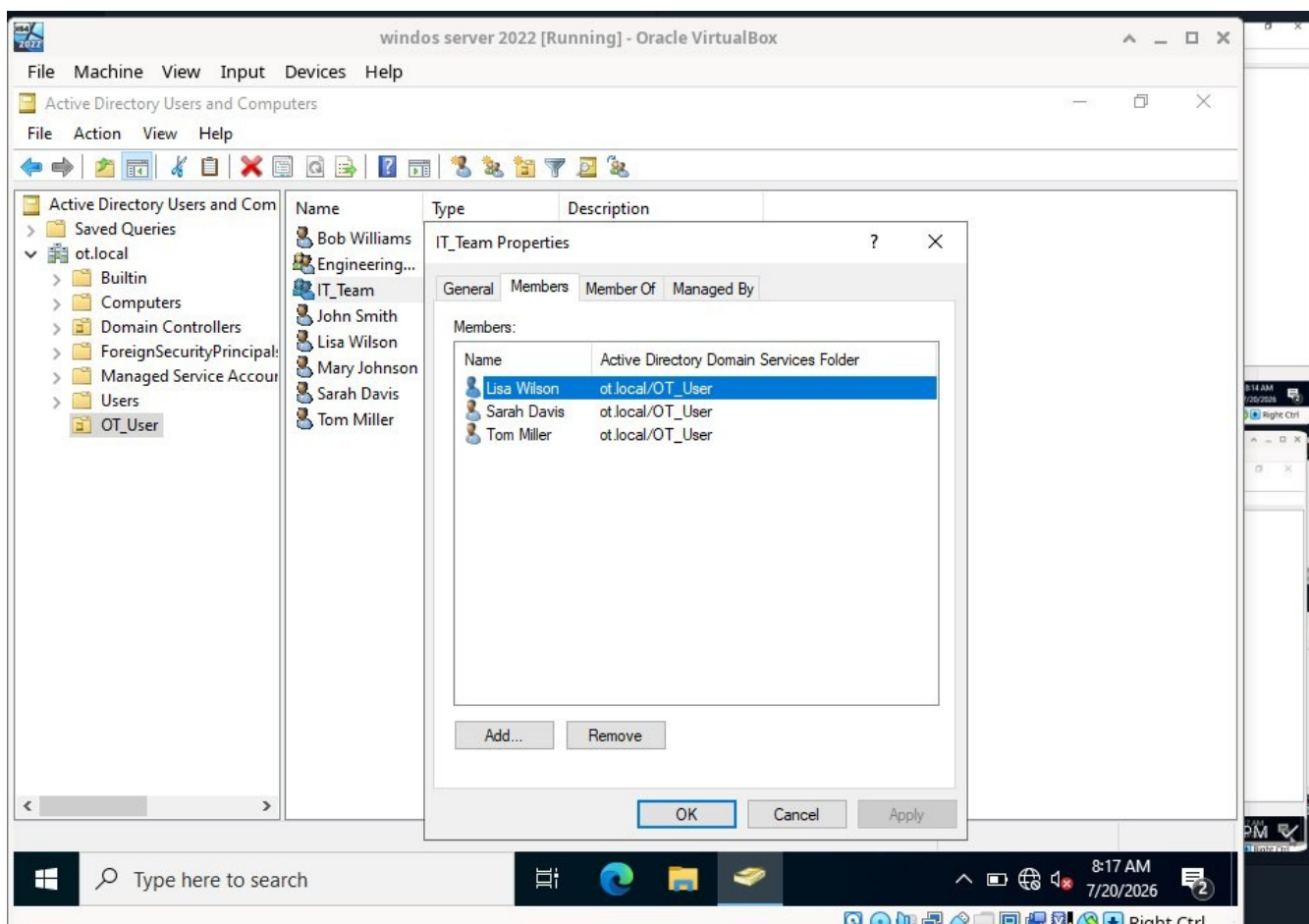
Engineering_Team Group Membership

IT Team

- sdavis
- tmiller
- lwilson

were added to:

IT_Team



IT_Team Group Membership]

User Account Summary

Full Name	Username	Assigned Security Group
John Smith	jsmith	Engineering_Team
Mary Johnson	mjohnson	Engineering_Team
Bob Williams	bwilliams	Engineering_Team
Sarah Davis	sdavis	IT_Team
Tom Miller	tmiller	IT_Team
Lisa Wilson	lwilson	IT_Team

Result

The Organizational Unit (**OT_Users**) was successfully created within the **ot.local** domain. Two Global Security Groups (**Engineering_Team** and **IT_Team**) were configured, and six user accounts were created and assigned to their appropriate groups. The completed directory structure provides centralized management of users and supports Role-Based Access Control (RBAC), allowing permissions to be managed efficiently through group membership rather than individual user assignments.

Discussion

Organizational Units provide a logical method for organizing Active Directory objects according to departments or administrative requirements. Separating users into dedicated Organizational Units simplifies management tasks such as delegation, policy application, and account administration.

Using Global Security Groups to manage user permissions implements the principles of Role-Based Access Control (RBAC). Rather than assigning permissions directly to individual users, permissions can be granted to security groups, and users inherit the appropriate access through their group membership. This approach reduces administrative overhead, improves consistency, and simplifies future maintenance as users join, leave, or change departments.

The completed Organizational Unit and security group structure establishes the foundation for subsequent exercises involving logon restrictions, account lifecycle management, file permissions, and access control testing.

Conclusion

The Active Directory organizational structure was successfully implemented by creating the **OT_Users** Organizational Unit, provisioning six user accounts, and organizing them into the **Engineering_Team** and **IT_Team** Global Security Groups. This centralized structure supports secure user administration and prepares the environment for implementing advanced security controls in the remaining exercises.

Exercise 3: Time-Based Logon Restrictions

Objective

The objective of this exercise was to strengthen account security by implementing time-based logon restrictions within Active Directory. By limiting user authentication to approved working hours, administrators can reduce the risk of unauthorized access outside normal business operations. This security control is particularly important in Operational Technology (OT) environments, where access to critical systems should be restricted to authorized personnel during designated operational periods.

Lab Execution Summary

This exercise focused on configuring logon hour restrictions for the user account **John Smith (jsmith)** using **Active Directory Users and Computers (ADUC)**. The account was configured to allow authentication only during normal business hours, specifically **Monday through Friday from 9:00 AM to 5:00 PM**. Any logon attempts outside this schedule would be denied by Active Directory.

Implementation Steps

Step 1: Open Active Directory Users and Computers

Logged into the Domain Controller (**DC01**) using the Domain Administrator account.

Opened **Server Manager**, selected **Tools**, and launched **Active Directory Users and Computers (ADUC)**.

Step 2: Locate the User Account

Expanded the **ot.local** domain and opened the **OT_Users** Organizational Unit.

Located the user account:

- **John Smith**
- **Username:** jsmith

Step 3: Open User Properties

Right-clicked the **jsmith** account and selected **Properties**.

Opened the **Account** tab to access the account configuration settings.

Step 4: Configure Logon Hours

Within the **Account** tab, selected **Logon Hours**.

The Logon Hours configuration window displayed the weekly schedule used to define permitted and denied logon periods.

John Smith Properties

General Address **Account** Profile Telephones Organization Mem

User logon name:
 @ot.local

User logon name (pre-Windows 2000):

☐ Unlock account

Account options:

- ☐ User must change password at next logon
- ☐ User cannot change password
- ☒ Password never expires
- ☐ Store password using reversible encryption

Account expires

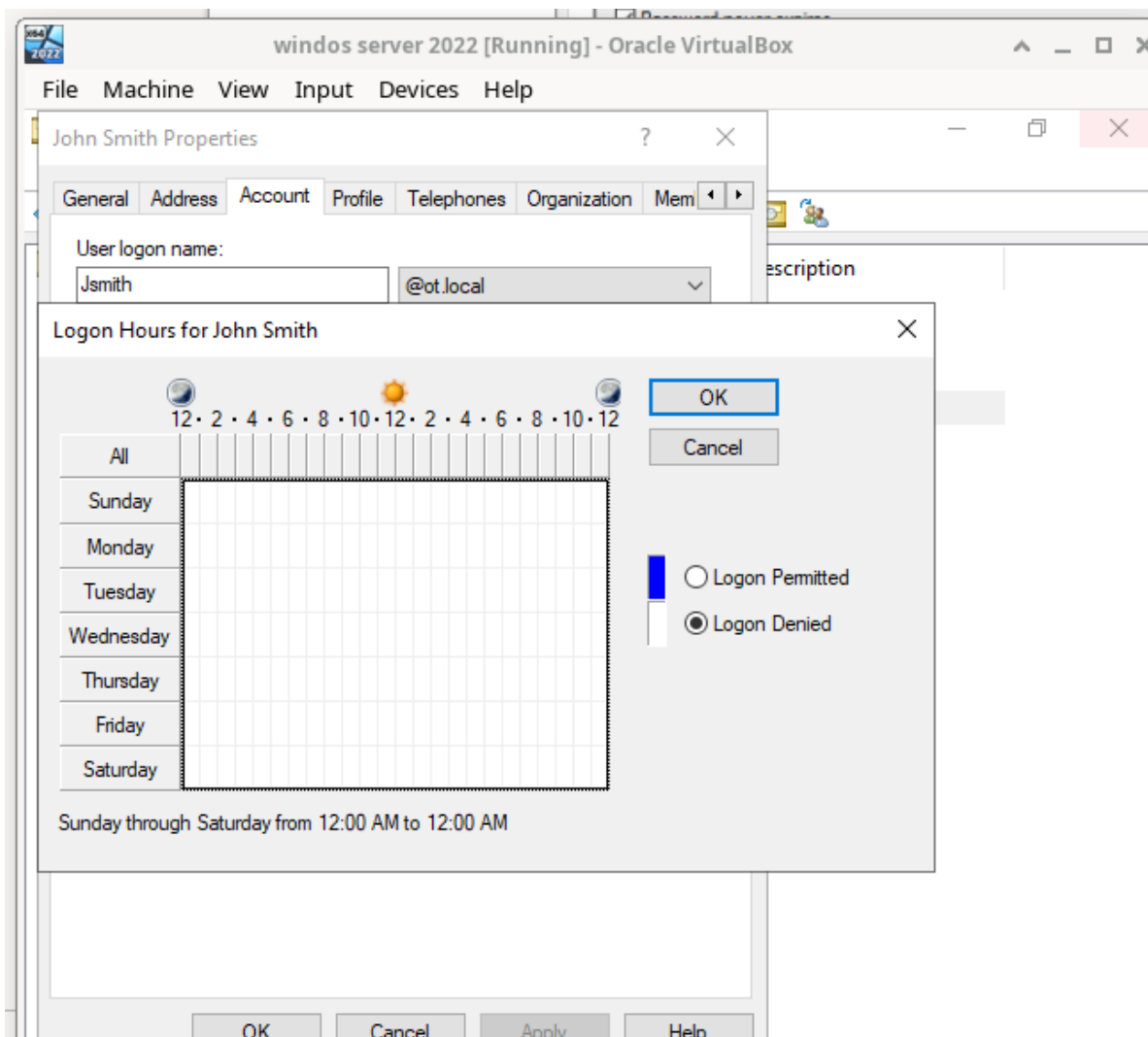
☒ Never

☐ End of:

Step 5: Restrict Access Outside Business Hours

Configured the logon schedule to permit authentication only during the following period:

Days	Allowed Time
Monday – Friday	9:00 AM – 5:00 PM
All remaining time periods, including evenings and weekends, were configured as Logon Denied .	



Step 6: Apply the Configuration

Selected **OK** to save the Logon Hours configuration.

Closed the user properties window and verified that the account settings were successfully updated.

Step 7: Verify the Restriction

Verified that the logon schedule had been applied successfully.

The configured account would now authenticate only during the permitted working hours. Any authentication attempts outside the configured schedule would be rejected by Active Directory.

Result

The user account **jsmith** was successfully configured with time-based logon restrictions. Authentication was permitted only from **Monday through Friday, 9:00 AM to 5:00 PM**, while all other time periods were denied. This configuration ensures that the account cannot be used outside approved working hours, improving security within the Active Directory environment.

Discussion

Time-based logon restrictions provide an additional layer of security by limiting when users are permitted to authenticate to the domain. Even if valid credentials are compromised, an attacker cannot successfully log in outside the authorized schedule.

In Operational Technology (OT) environments, this control helps protect industrial systems by ensuring that administrative and engineering accounts are used only during approved operational periods. Restricting after-hours access also reduces the likelihood of unauthorized system modifications and supports organizational security policies.

Conclusion

The logon hours policy for **jsmith** was successfully implemented using Active Directory Users and Computers. By allowing authentication only during standard business hours (Monday through Friday, 9:00 AM to 5:00 PM), the exercise demonstrated how Active Directory can enforce time-based access control as part of a defense-in-depth security strategy for Operational Technology environments.

Exercise 4: Account Lifecycle Management

Objective

The objective of this exercise was to demonstrate how Active Directory manages the lifecycle of user accounts through administrative account controls. The exercise focused on two common account

management tasks: disabling user accounts for inactive or departing employees and enforcing password changes at the next logon. These controls help maintain the security of the domain by preventing unauthorized access and ensuring that user credentials remain secure.

Lab Execution Summary

This exercise simulated two real-world administrative scenarios within an Operational Technology (OT) environment:

- **Disabling the account** of an inactive employee (mjohanson) to immediately prevent authentication.
- **Requiring a password change at the next logon** for another user (bwilliams) to enforce secure password management.

Both tasks were performed using **Active Directory Users and Computers (ADUC)** to demonstrate standard account lifecycle management practices.

Implementation Steps

Part A: Disabling a User Account

Step 1: Open Active Directory Users and Computers

Logged into the Domain Controller (**DC01**) using the Domain Administrator account.

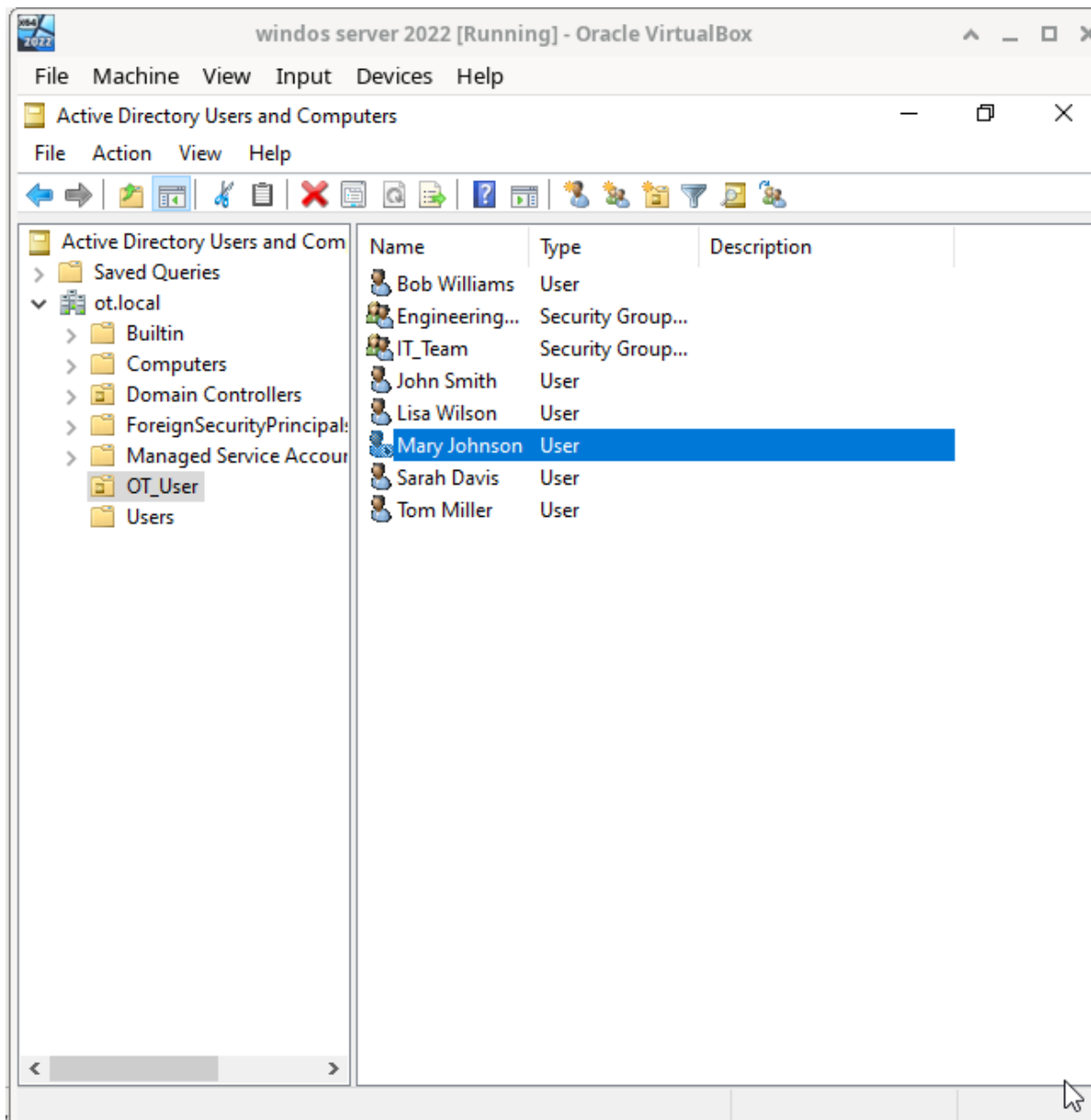
Opened **Server Manager**, selected **Tools**, and launched **Active Directory Users and Computers (ADUC)**.

Step 2: Locate the User Account

Expanded the **ot.local** domain and opened the **OT_Users** Organizational Unit.

Located the user account:

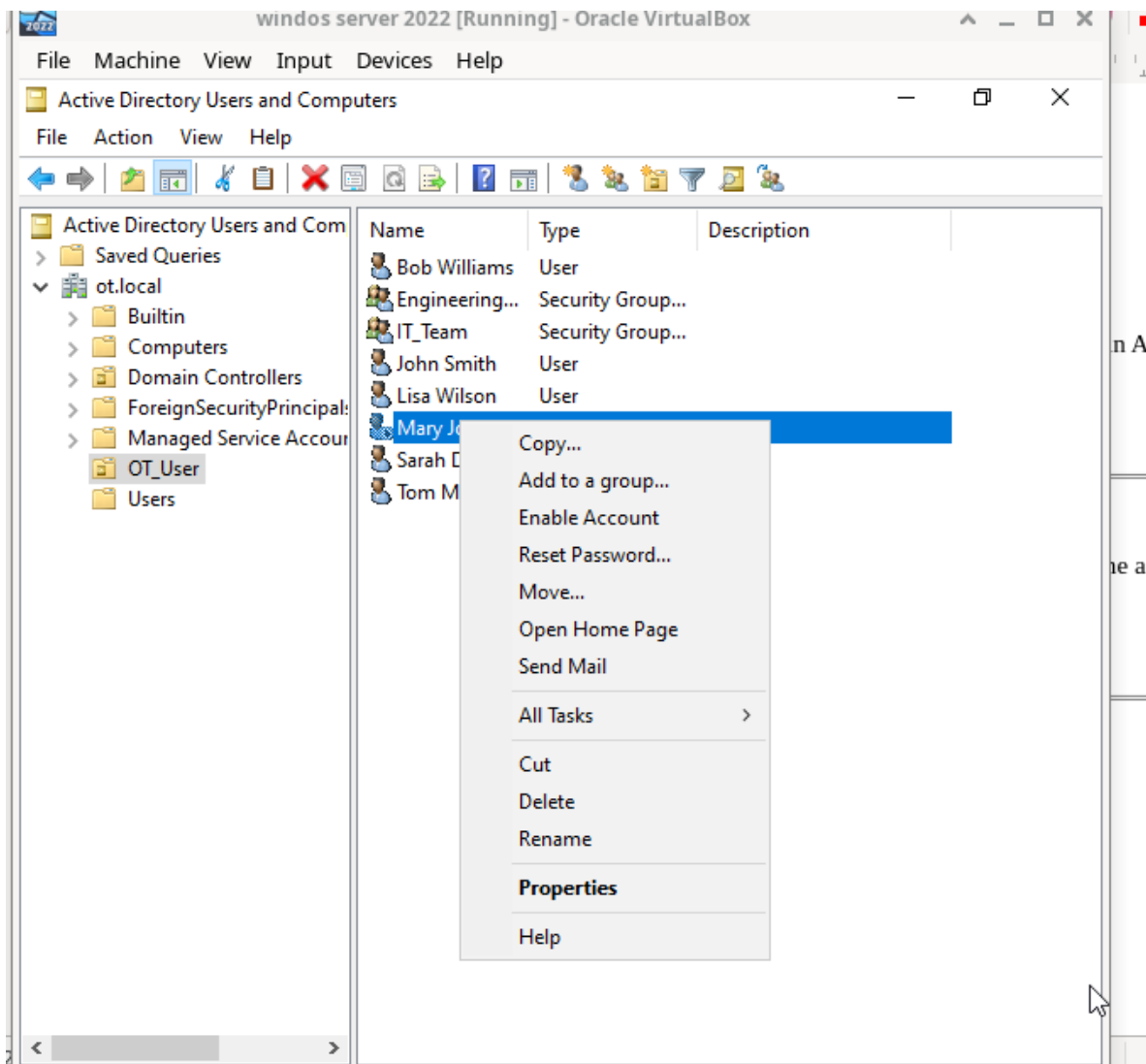
- **Mary Johnson**
- **Username:** mjohanson



Step 3: Disable the Account

Right-clicked the **mjohnson** user account and selected **Disable Account**.

The account status changed immediately, displaying the disabled account indicator in Active Directory Users and Computers.



Step 4: Verify the Account Status

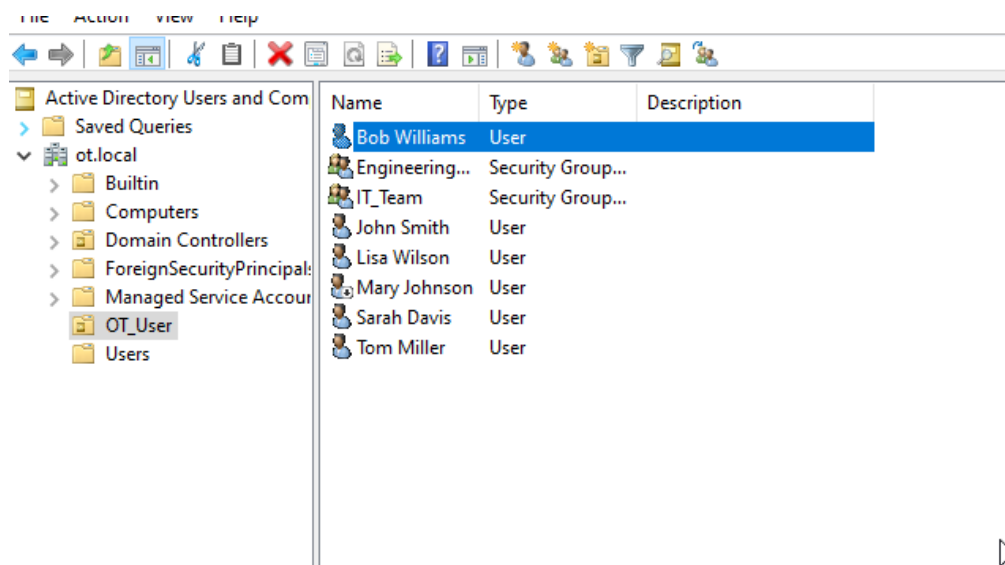
Confirmed that the **mjohnson** account displayed the disabled icon, indicating that the account could no longer authenticate to the domain.

Part B: Force Password Change at Next Logon

Step 5: Locate the User Account

Within the **OT_Users** Organizational Unit, located the following user account:

- **Bob Williams**
- **Username: bwilliams**



Step 6: Open Account Properties

Right-clicked the **bwilliams** account and selected **Properties**.

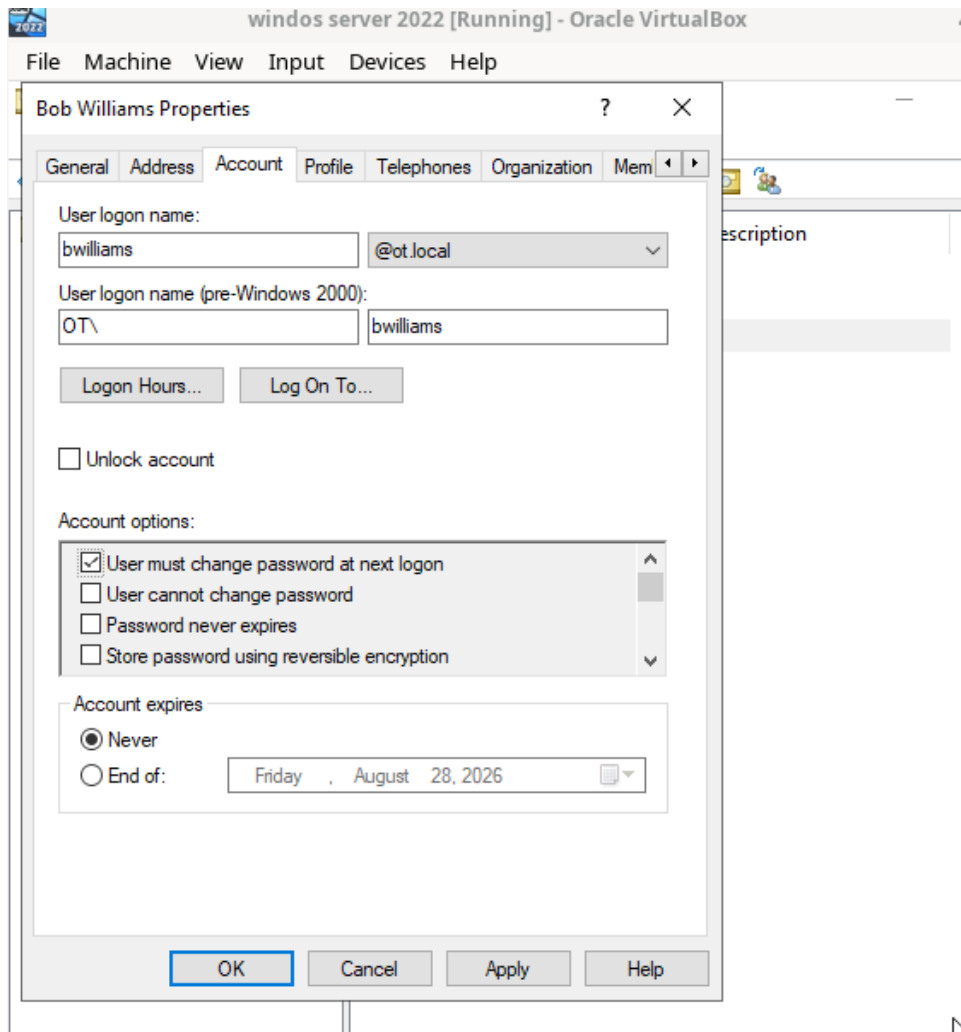
Opened the **Account** tab to modify the account settings.

Step 7: Configure Password Settings

Enabled the option:

- **User must change password at next logon**

This configuration ensures that the user is prompted to create a new password during the next successful authentication.



Step 8: Apply the Configuration

Selected **Apply** and **OK** to save the account configuration.

Verified that the password policy had been successfully applied.

Result

The account lifecycle management tasks were completed successfully. The **mjohnson** account was disabled, immediately preventing future authentication attempts, while the **bwilliams** account was configured to require a password change during the next logon. These administrative controls demonstrate how Active Directory can be used to manage user account status and enforce password security throughout the account lifecycle.

Discussion

Account lifecycle management is an essential responsibility of Active Directory administrators. Disabling inactive accounts helps prevent unauthorized access using credentials that are no longer required, while preserving the account for future reactivation if necessary.

Requiring users to change their password at the next logon strengthens account security by ensuring that temporary or administrator-assigned passwords are replaced with user-specific credentials. This reduces the risk of compromised default passwords and supports organizational password management policies.

Within an Operational Technology (OT) environment, these controls help protect critical infrastructure by ensuring that only authorized personnel maintain active access to industrial systems.

Conclusion

The account lifecycle management exercise successfully demonstrated two important administrative security controls within Active Directory. The **mjohnson** account was disabled to simulate the removal of access for an inactive employee, while the **bwilliams** account was configured to require a password change at the next logon. These controls enhance domain security, support proper user account management, and contribute to maintaining a secure Operational Technology environment.

Exercise 5: Granular Network Share and NTFS Permissions

Objective

The objective of this exercise was to configure secure network file sharing by implementing granular NTFS permissions and enabling Access-Based Enumeration (ABE). The exercise demonstrates how Role-Based Access Control (RBAC), explicit NTFS permissions, and Access-Based Enumeration work together to ensure that users can access only the files and folders required for their job responsibilities within an Operational Technology (OT) environment.

Lab Execution Summary

In this exercise, a shared folder named **Staff** was created on the Domain Controller to serve as a centralized departmental file repository. Two departmental folders (**Engineering** and **IT**) were configured with separate NTFS permissions based on security group membership. Permission inheritance was modified, explicit Allow and Deny permissions were applied, and Access-Based Enumeration (ABE) was enabled to hide folders from unauthorized users. This configuration demonstrates the Principle of Least Privilege (PoLP) by ensuring users have access only to resources required for their assigned roles.

Implementation Steps

Step 1: Create the Staff Folder

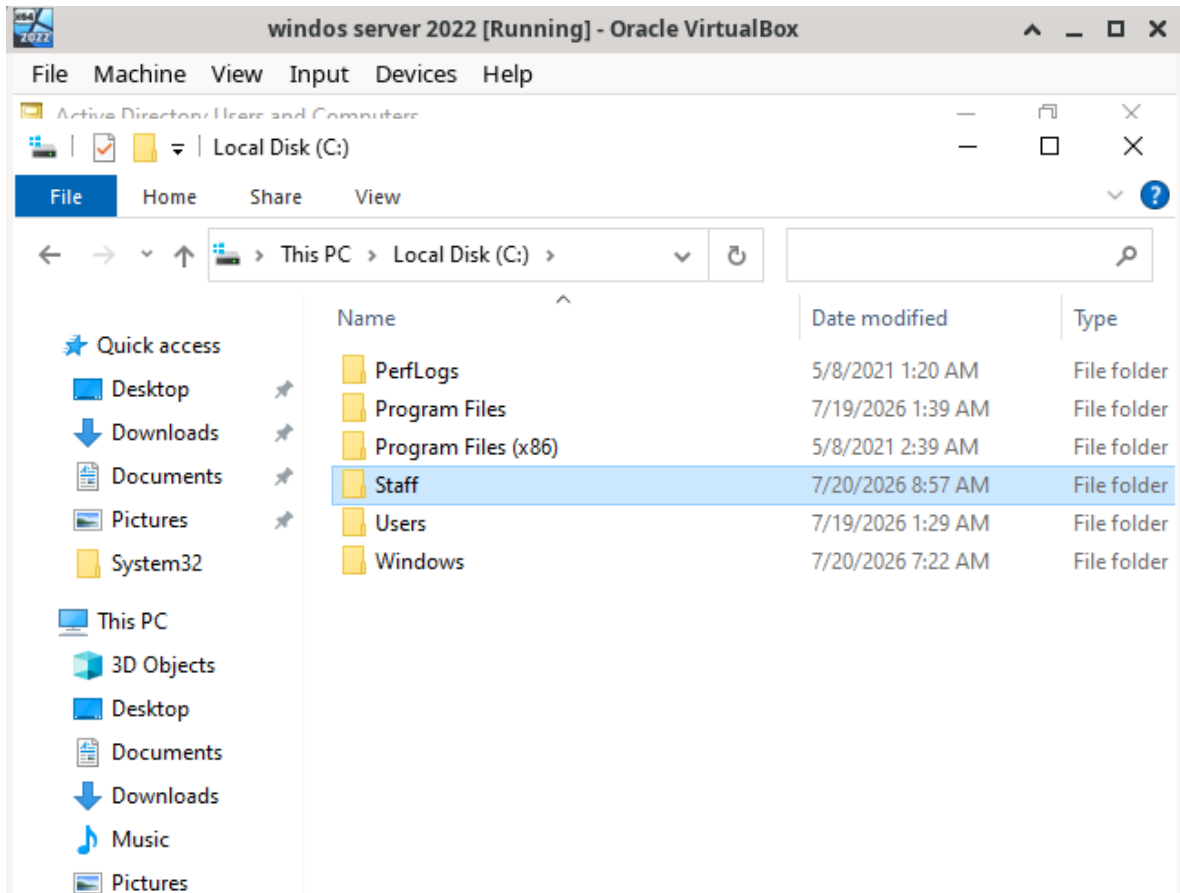
Logged into the Domain Controller (**DC01**) using the Domain Administrator account.

Opened **File Explorer** and navigated to the **C:** drive.

Created a new folder named:

C:\Staff

This folder serves as the main shared directory for departmental resources.



Step 2: Create Departmental Folders

Inside the **Staff** folder, created two subfolders:

- **Engineering**

- **IT**

These folders will store files specific to each department.

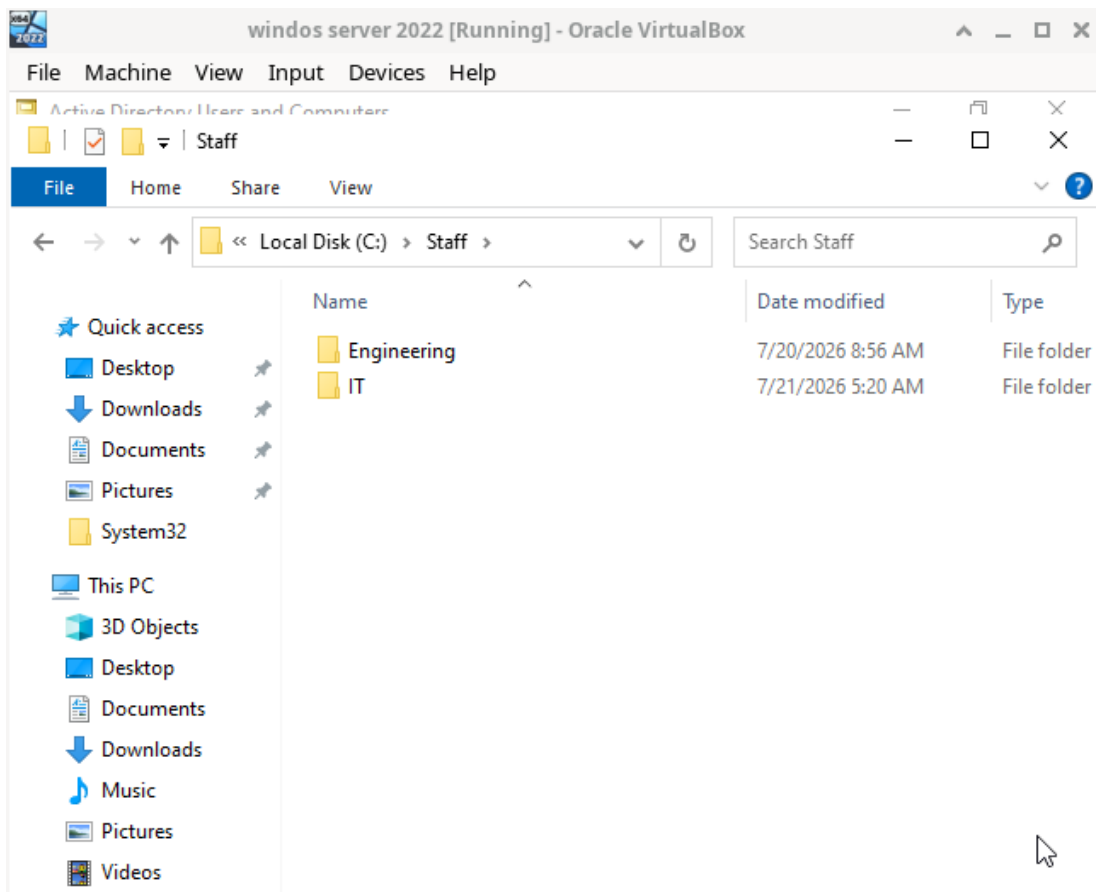
Folder structure:

C:\

└─ Staff

 └─ Engineering

 └─ IT



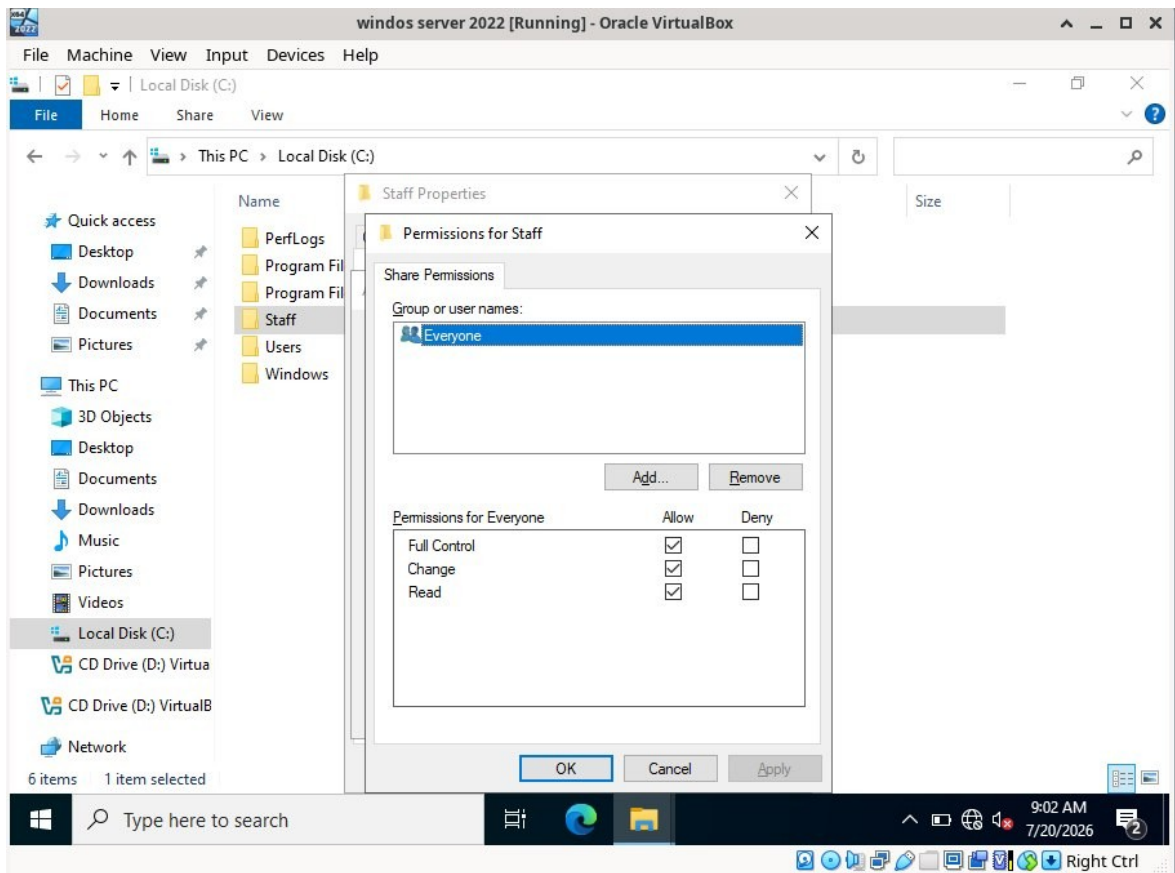
Step 3: Configure the Network Share

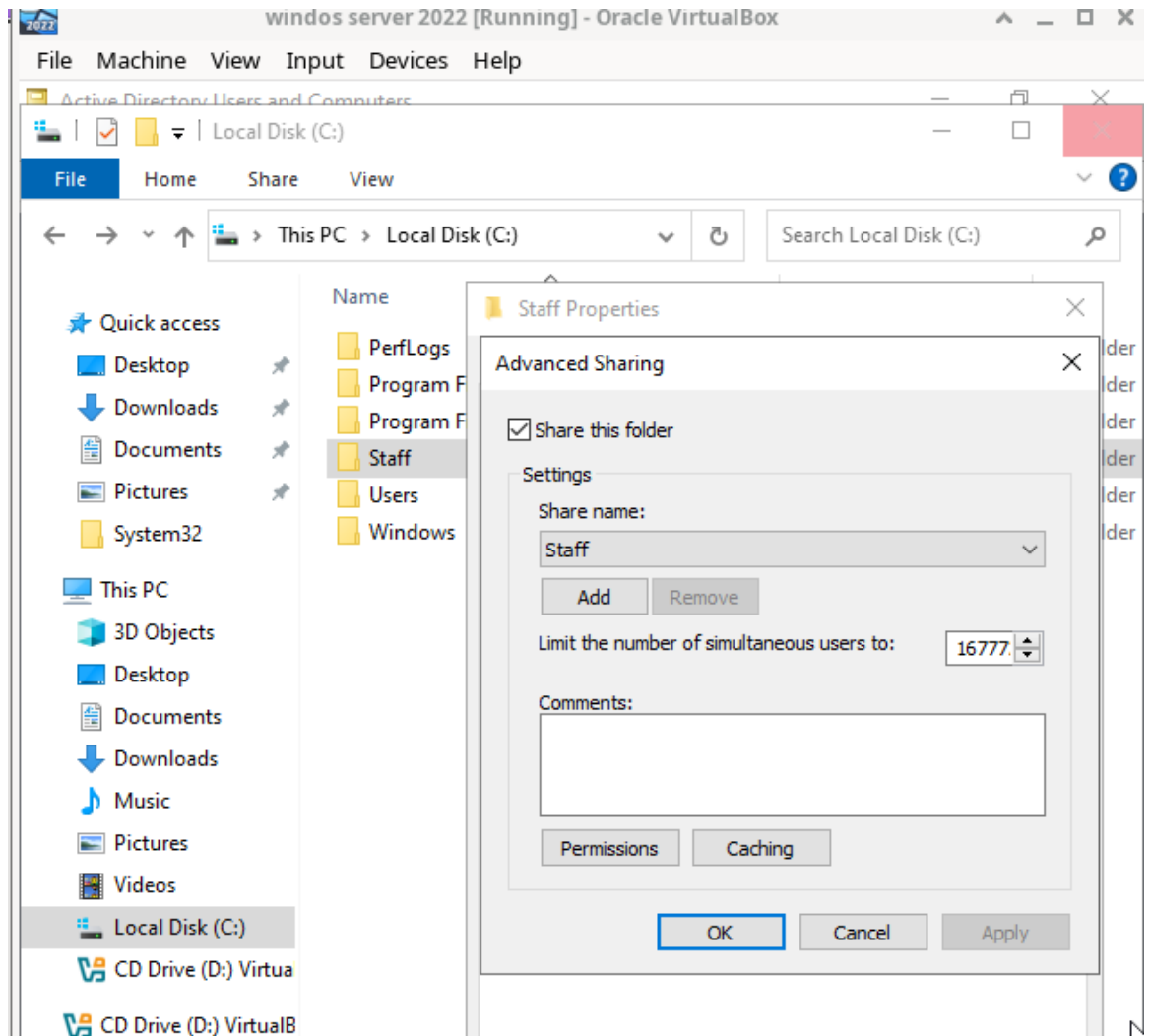
Right-clicked the **Staff** folder and selected **Properties**.

Opened the **Sharing** tab and selected **Advanced Sharing**.

Performed the following actions:

- Enabled **Share this folder**
- Assigned the share name **Staff**
- Configured the required share permissions according to the lab requirements.





Step 4: Configure NTFS Permissions for the Engineering Folder

Opened the **Properties** of the **Engineering** folder.

Selected:

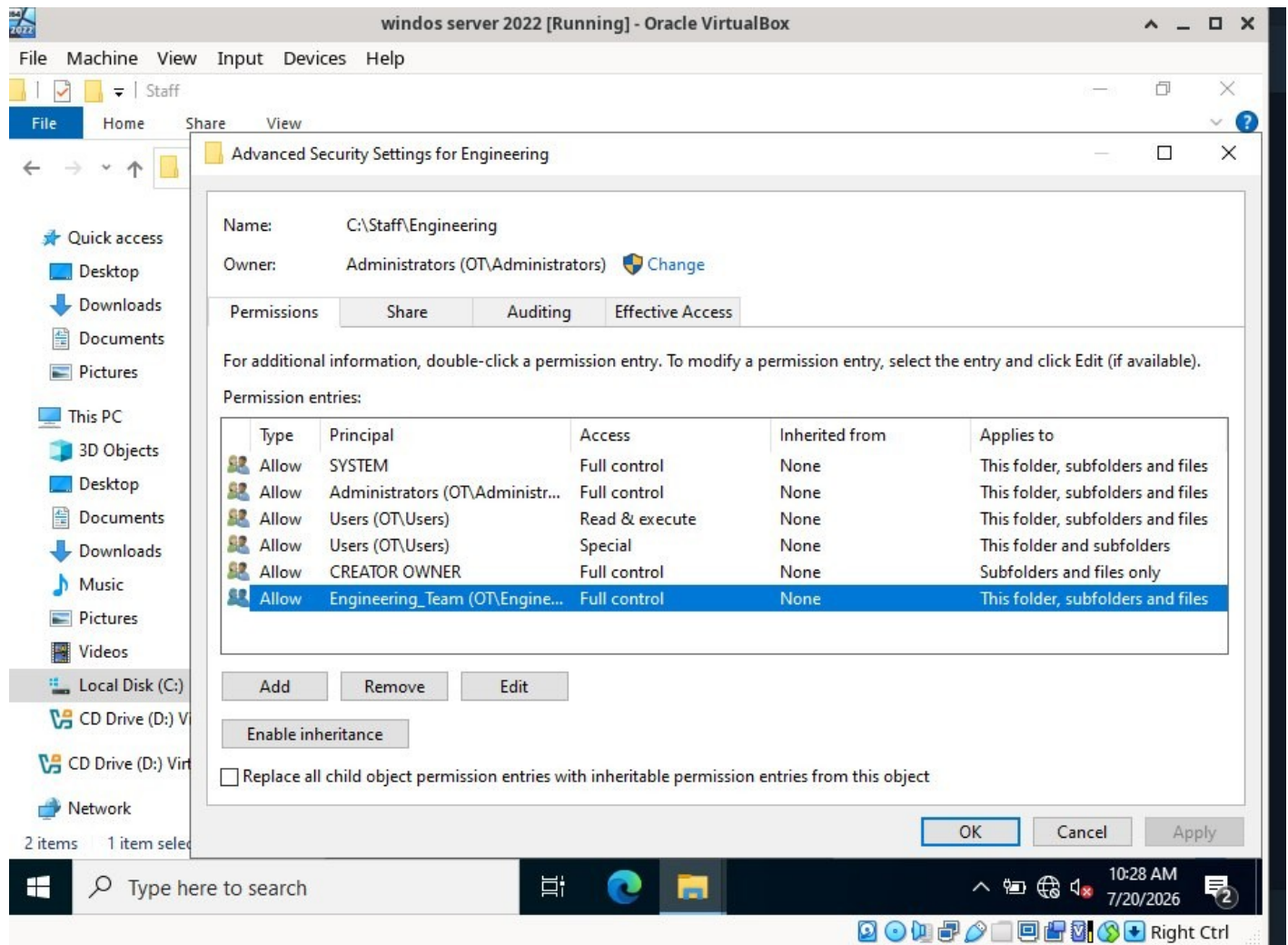
Security → **Advanced**

Performed the following configuration:

- Disabled permission inheritance.
- Removed inherited permissions from the parent folder.

- Added **Engineering_Team** with **Modify** permissions.
- Added **IT_Team** with an **Explicit Deny** permission to prevent access.

This configuration ensures that only Engineering personnel can access the Engineering directory.



Step 5: Configure NTFS Permissions for the IT Folder

Opened the **Properties** of the **IT** folder.

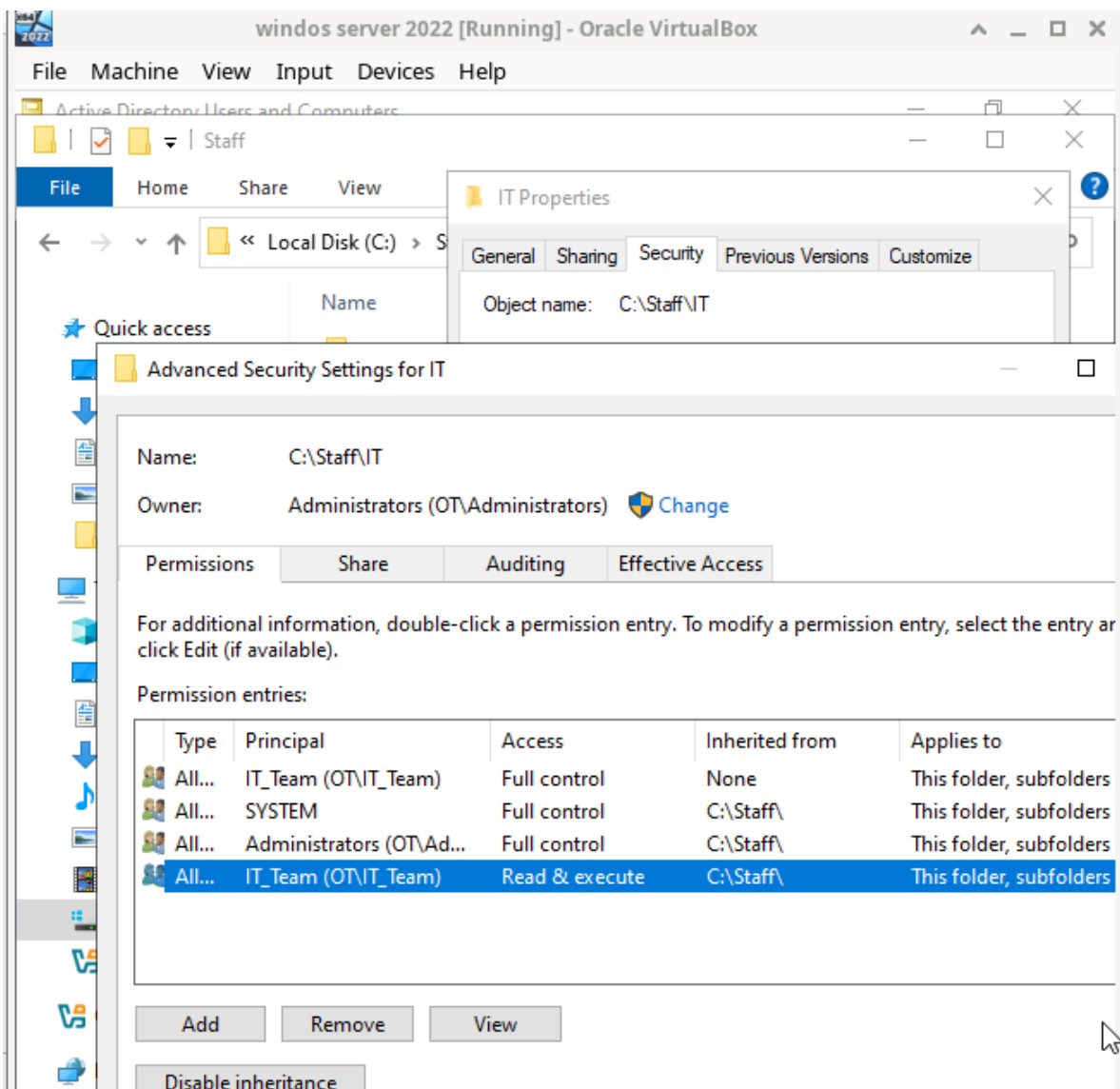
Selected:

Security → **Advanced**

Performed the following configuration:

- Disabled permission inheritance.
- Converted inherited permissions where required.
- Granted **IT_Team Modify** permissions.
- Removed unnecessary permissions so that Engineering users could not access the folder.

This configuration ensures that only IT personnel can access the IT directory.



Step 6: Enable Access-Based Enumeration (ABE)

Opened **Server Manager**.

Navigated to:

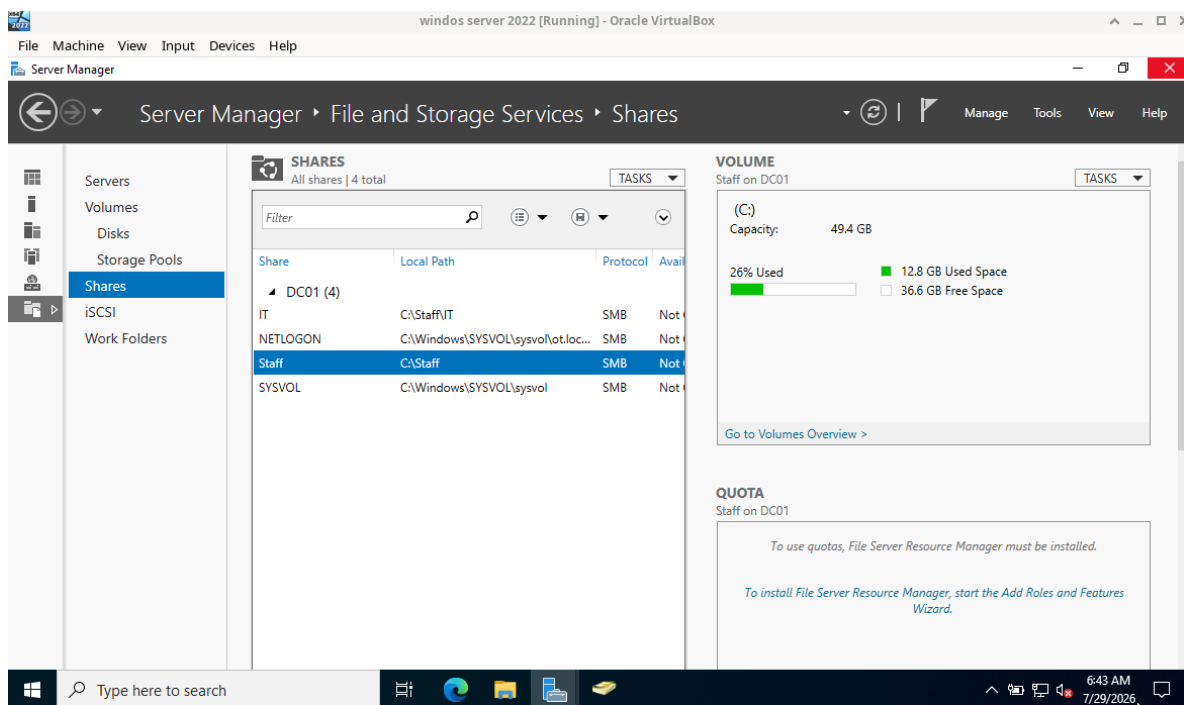
File and Storage Services → Shares

Selected the **Staff** share and opened its **Properties**.

Enabled:

Access-Based Enumeration (ABE)

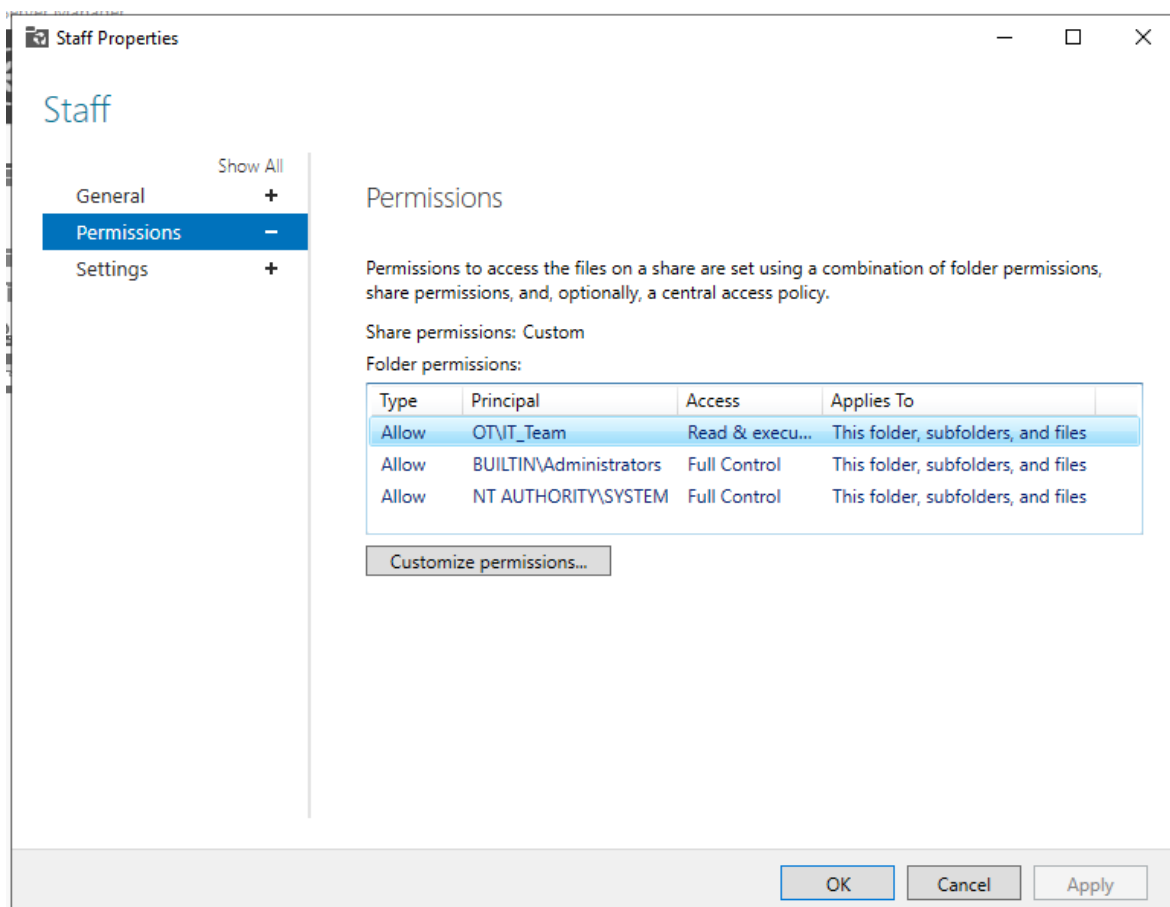
This configuration hides folders from users who do not have permission to access them.



Step 7: Verify Share Configuration

Reviewed the share configuration to verify that:

- The **Staff** folder was successfully shared.
- Engineering and IT folders had separate NTFS permissions.
- Access-Based Enumeration was enabled.
- Security groups were assigned the appropriate permissions.



Permission Configuration Summary

Folder	Security Group	Permission
Engineering	Engineering_Team	Modify (Allow)
Engineering	IT_Team	Explicit Deny
IT	IT_Team	Modify (Allow)
IT	Engineering_Team	No access

Result

The shared **Staff** directory was successfully created and configured. Separate departmental folders were protected using NTFS permissions based on Active Directory security groups. Access-Based Enumeration was enabled so that users could view only the folders they were authorized to access. This configuration successfully implemented Role-Based Access Control (RBAC) and the Principle of Least Privilege within the Active Directory environment.

Discussion

Properly configuring NTFS permissions is essential for protecting sensitive organizational data. Rather than granting permissions directly to individual users, permissions were assigned to Active Directory security groups. This simplifies administration and ensures consistent access control across the organization.

Disabling permission inheritance allowed each departmental folder to have independent security settings. The **Engineering** folder explicitly granted access to **Engineering_Team** while explicitly denying access to **IT_Team**, demonstrating the precedence of Explicit Deny permissions. Similarly, the **IT** folder was configured so that only members of **IT_Team** could access its contents.

Enabling Access-Based Enumeration (ABE) further strengthened security by preventing unauthorized users from seeing folders for which they lacked Read permissions. Instead of receiving an "Access Denied" message after selecting a folder, users simply do not see folders outside their authorization, reducing unnecessary information disclosure and limiting opportunities for internal reconnaissance.

Conclusion

The network file sharing environment was successfully secured using NTFS permissions, Active Directory security groups, and Access-Based Enumeration. Departmental folders were isolated according to job responsibilities, ensuring that Engineering users could access only Engineering resources and IT users could access only IT resources. The implementation demonstrated effective Role-Based Access Control (RBAC), the Principle of Least Privilege, and secure file-sharing practices appropriate for an Operational Technology (OT) environment.

Exercise 6: Access Control and Security Testing

Objective

The objective of this exercise was to verify that the Active Directory environment and security controls were functioning correctly by joining a client workstation to the **ot.local** domain and testing user authentication, account restrictions, and file access permissions. The exercise validated the implementation of Role-Based Access Control (RBAC), Access-Based Enumeration (ABE), NTFS permissions, account lifecycle management, and time-based logon restrictions.

Lab Execution Summary

After configuring the Domain Controller and implementing security policies, a client workstation (**CLIENT01**) was connected to the **ot.local** domain. The client was configured to use the Domain Controller as its DNS server, allowing communication with Active Directory. Various domain user accounts were then used to verify authentication, account restrictions, and departmental file access. The results confirmed that Active Directory policies and file permissions were operating as expected.

Implementation Steps

Step 1: Configure the Client Network Settings

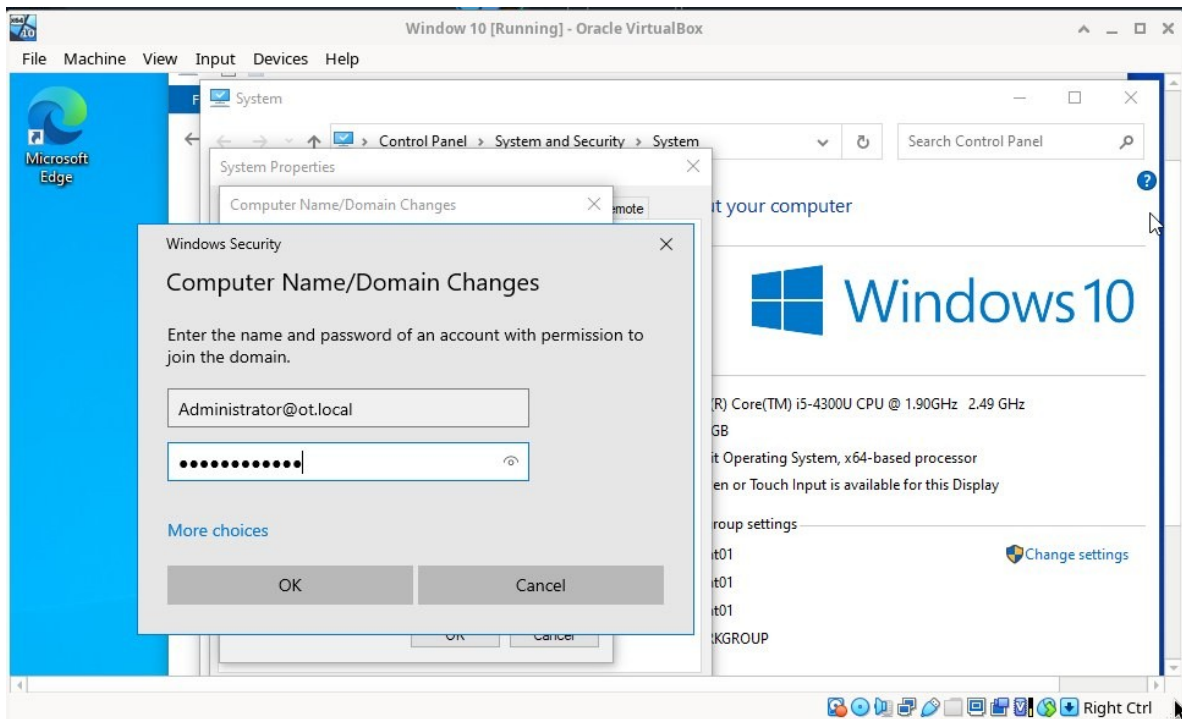
Logged into the client workstation (**CLIENT01**).

Opened the IPv4 network configuration and verified that the preferred DNS server was configured to use the Domain Controller.

Preferred DNS Server:

- **192.168.1.200**

This allows the client to locate and communicate with the Active Directory Domain Controller.



Step 2: Join the Client to the Domain

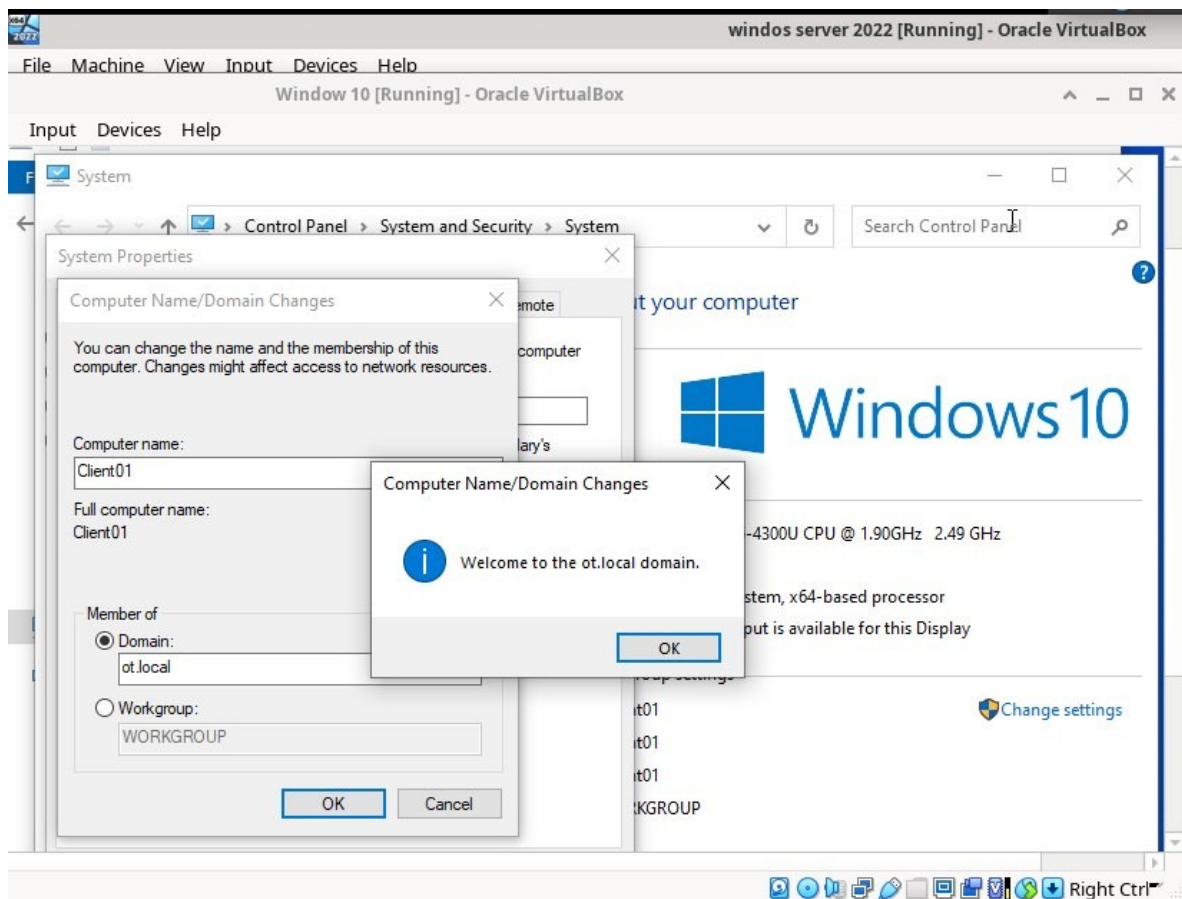
Opened:

System Properties → **Computer Name**

Selected **Change**, changed the membership type from **Workgroup** to **Domain**, and entered:

ot.local

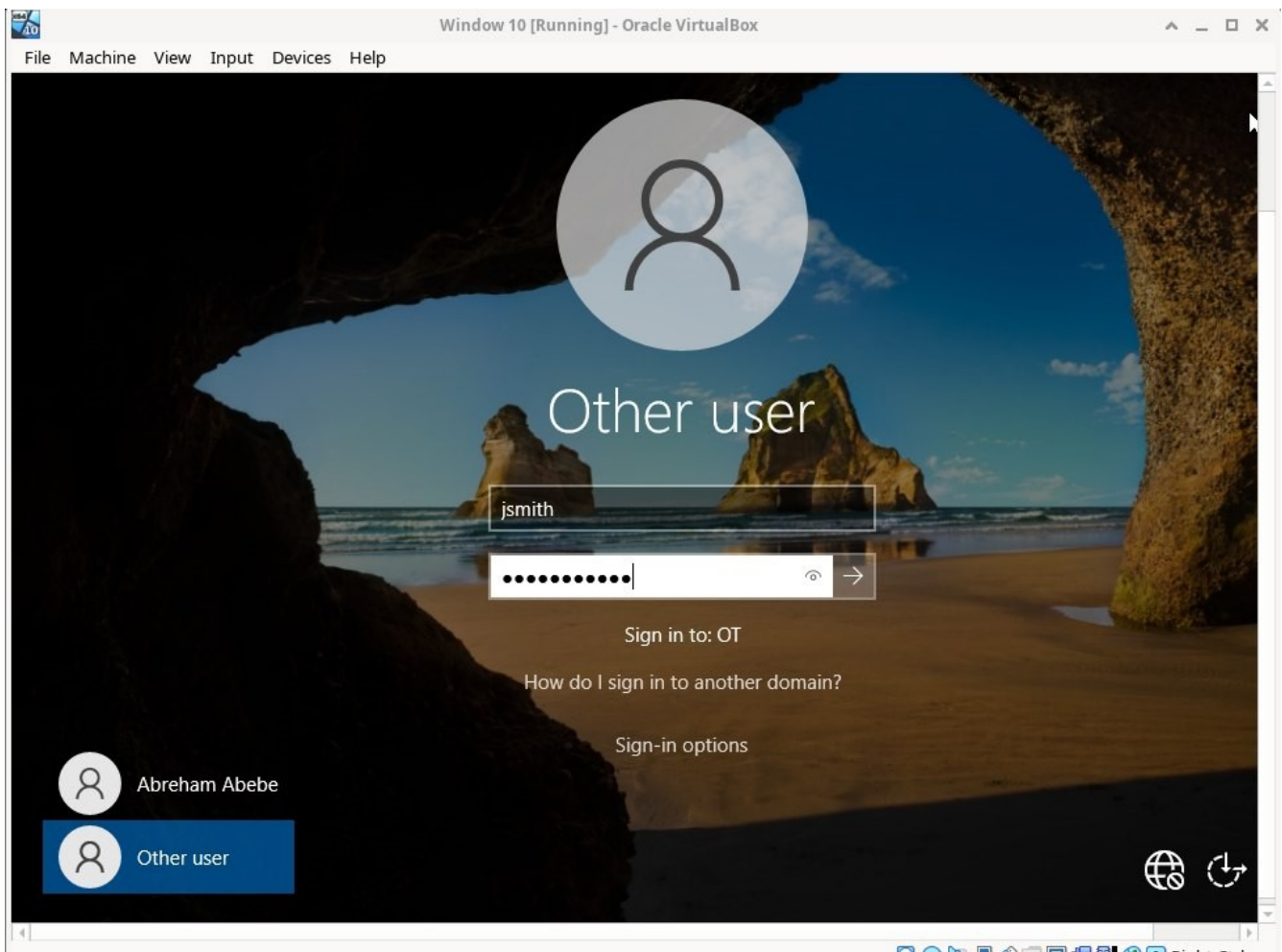
Authenticated using Domain Administrator credentials and completed the domain join process.



Step 3: Restart the Client Workstation

Restarted **CLIENT01** to complete the domain join.

After rebooting, verified that the workstation was successfully joined to the **ot.local** domain.



Step 4: Test Time-Based Logon Restrictions

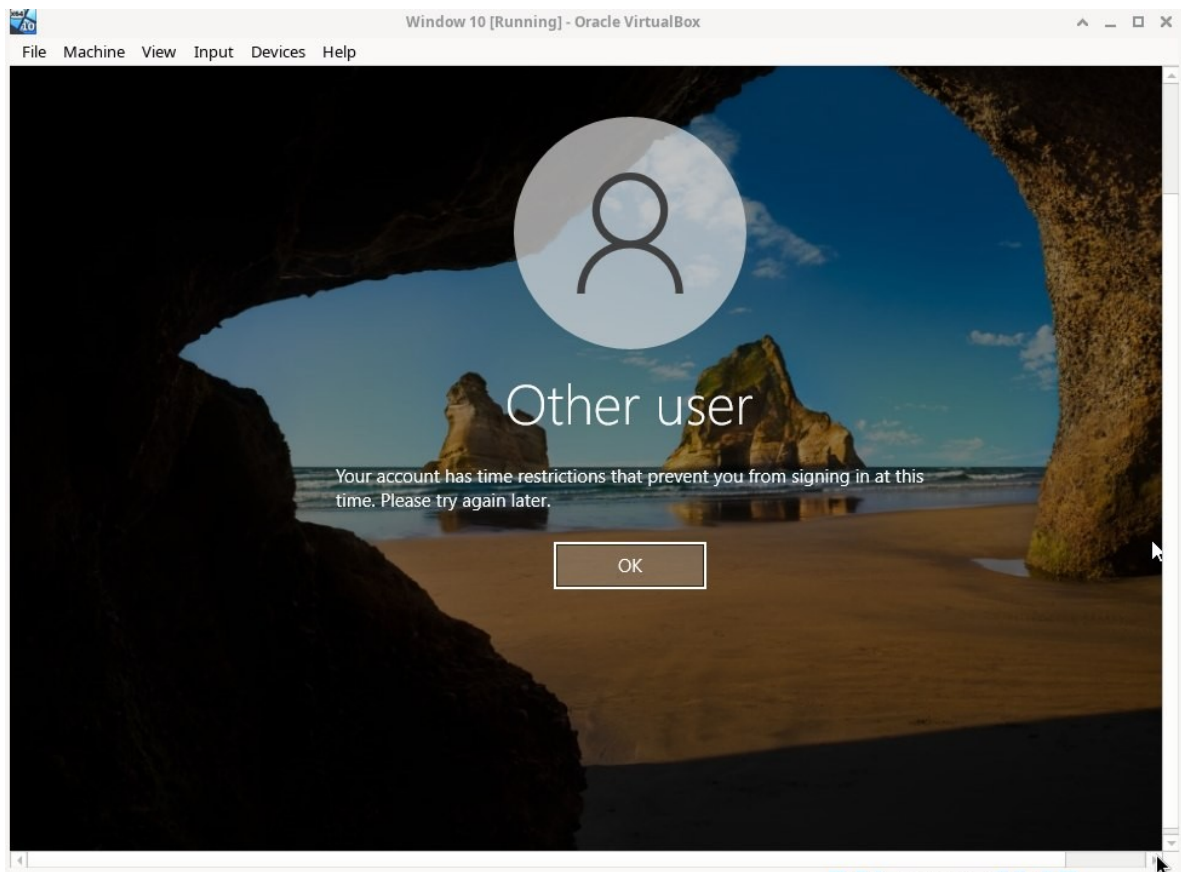
Logged into **CLIENT01** using the following account:

- **Username:** jsmith

Verified that:

- Logon was permitted only during the configured working hours.
- Authentication attempts outside the permitted schedule were denied.

This confirmed that the configured logon hours policy was successfully enforced by Active Directory.

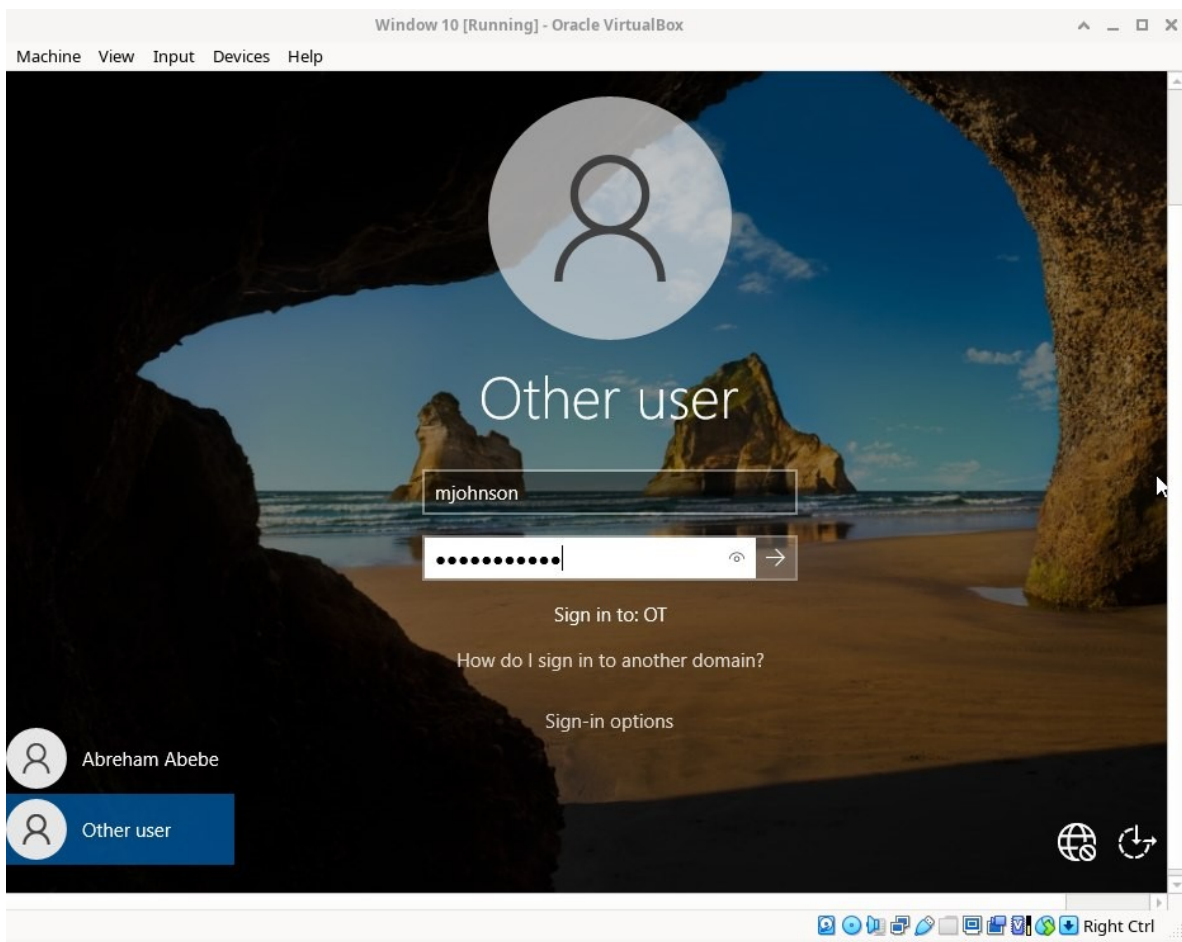


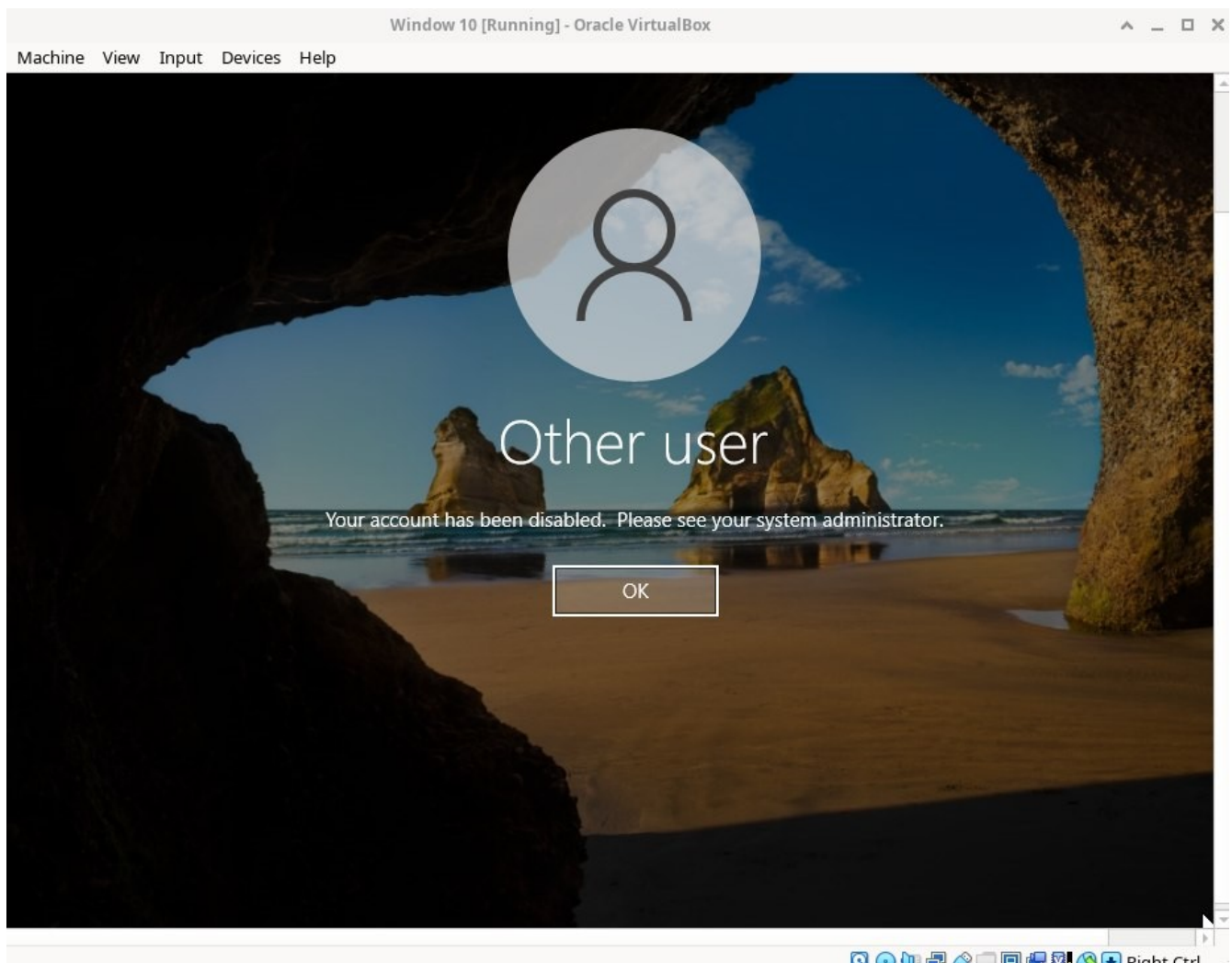
Step 5: Test the Disabled Account

Attempted to log in using:

- **Username:** mjohnson

Verified that authentication was denied because the account had been disabled during Exercise 4.





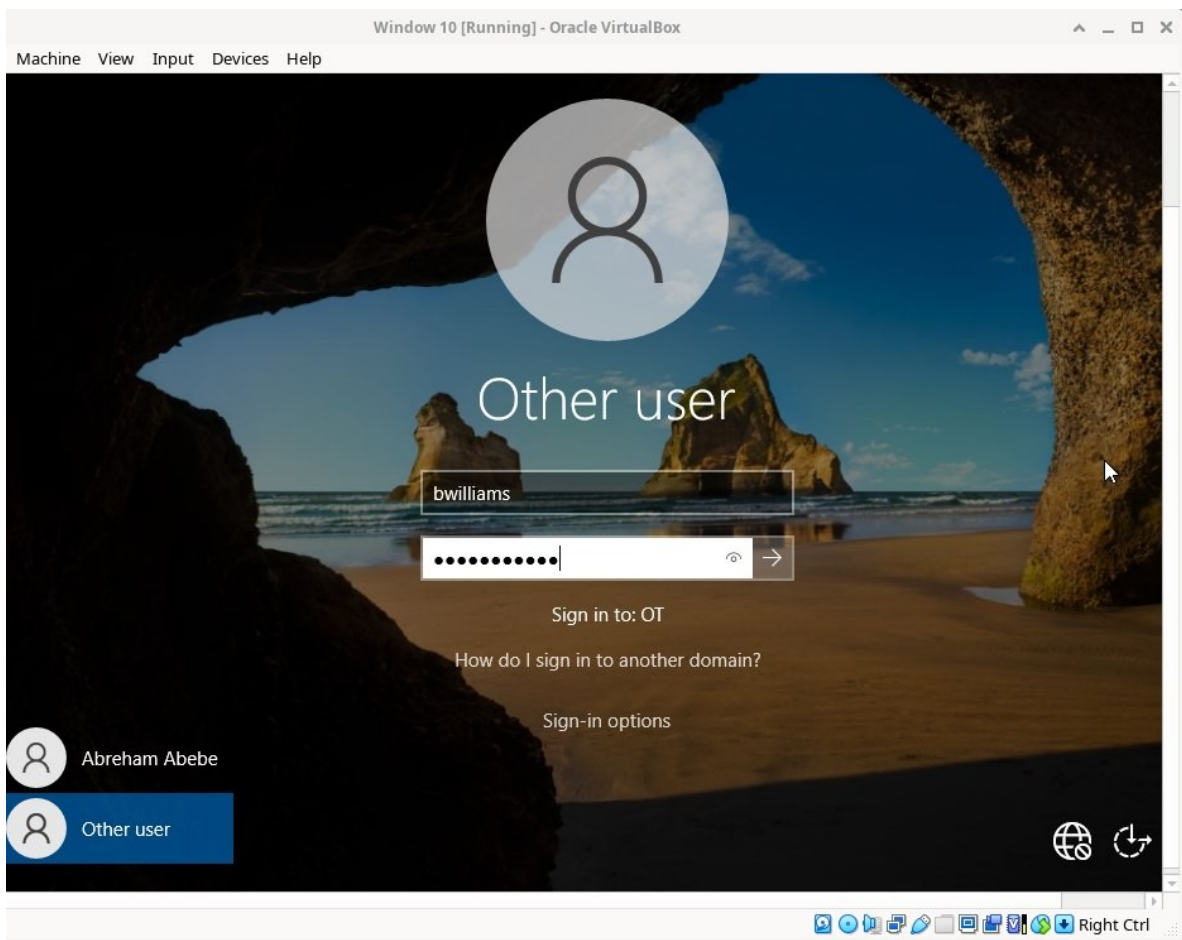
Step 6: Test Password Change Requirement

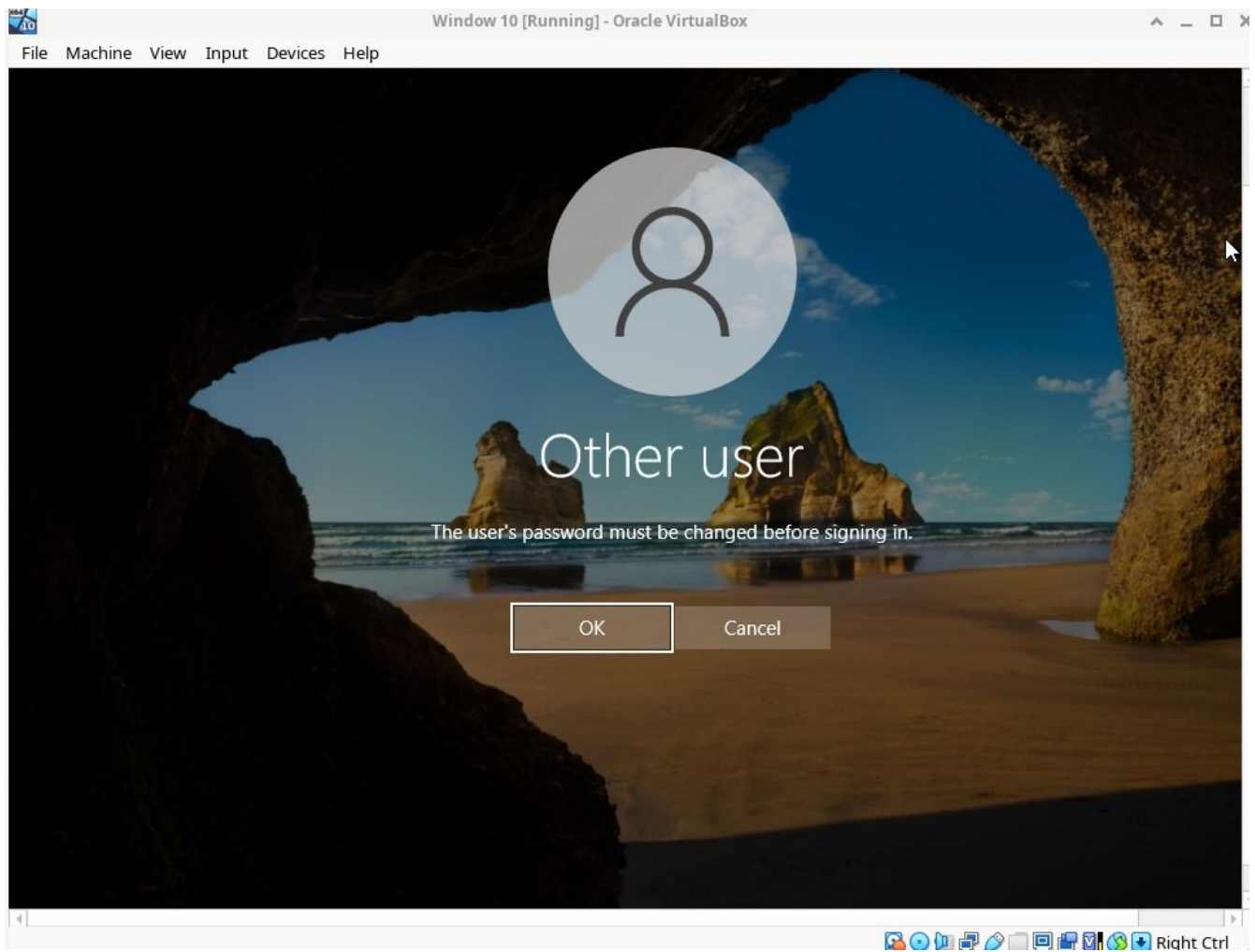
Logged in using:

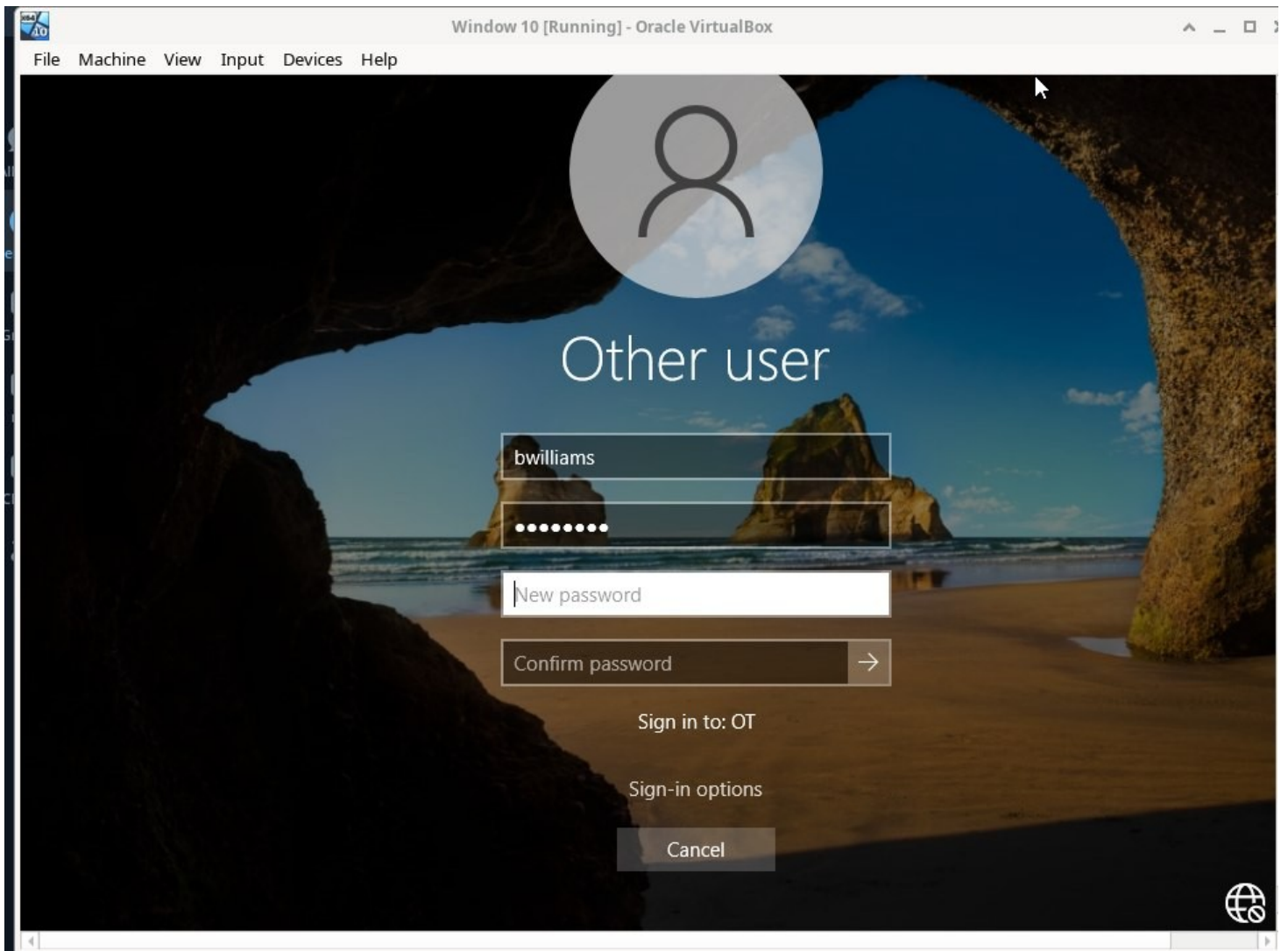
- **Username:** bwilliams

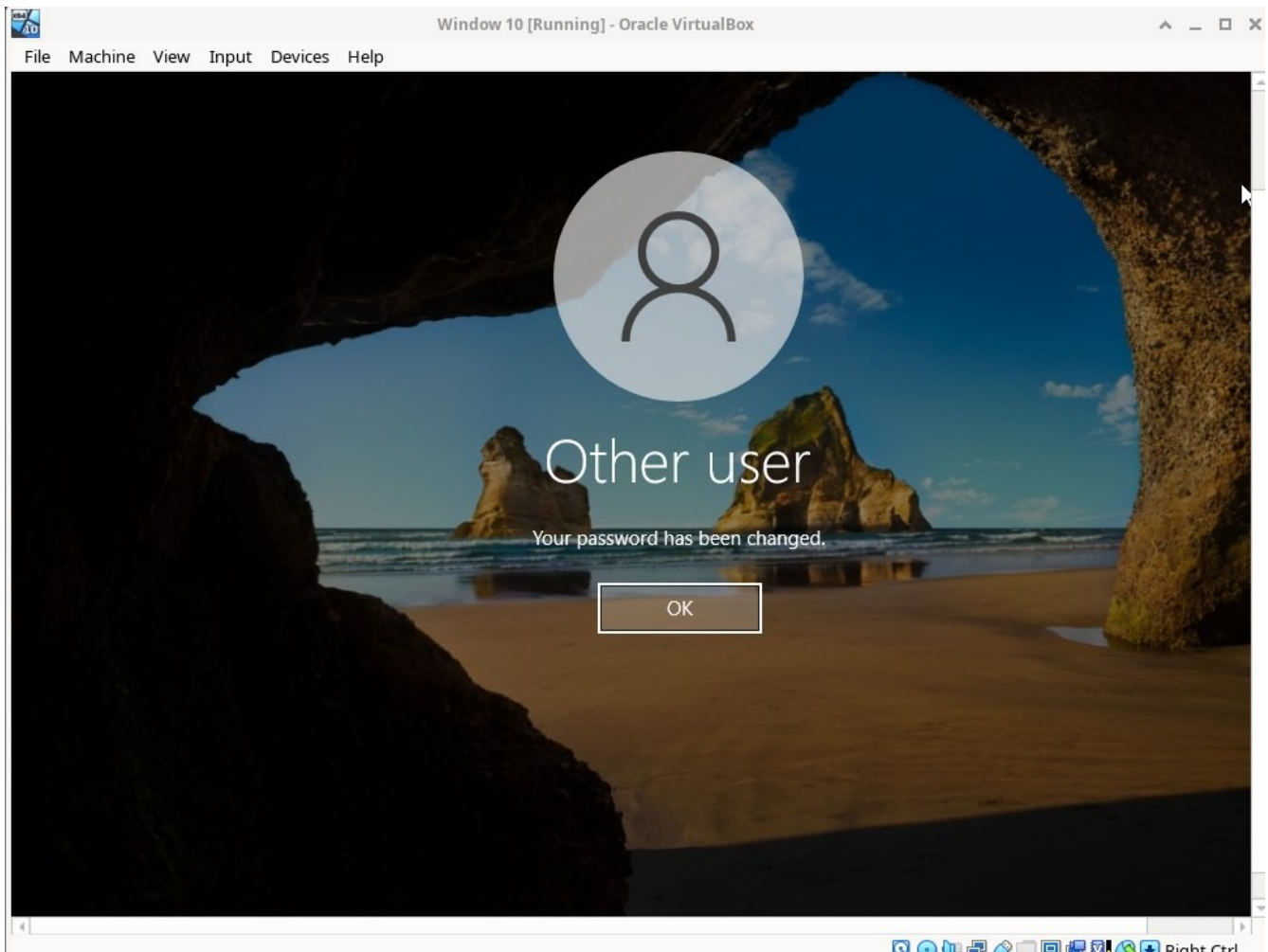
Verified that Windows prompted the user to change the password before completing the logon process.

After changing the password, the user successfully accessed the domain.









Step 7: Verify Engineering Department Access

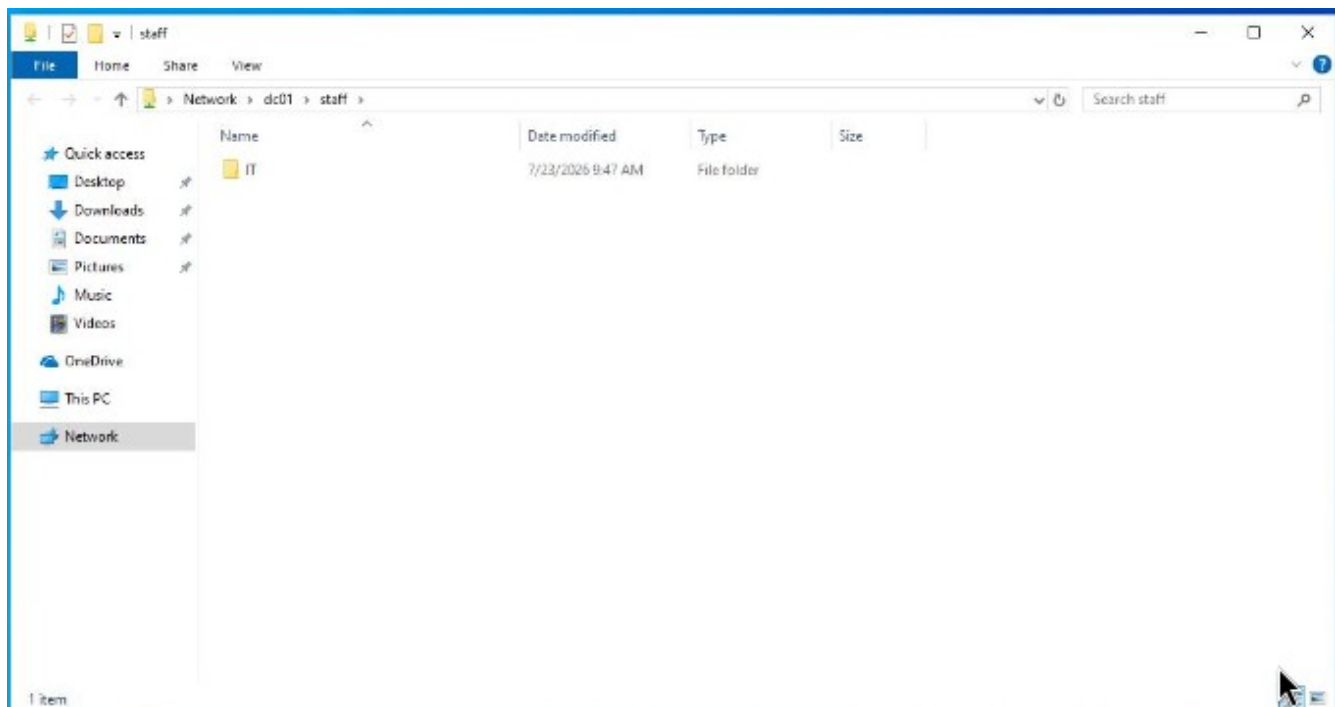
Logged into **CLIENT01** using the **jsmith** account.

Opened the shared folder:

\\DC01\\Staff

Verified that:

- The **Engineering** folder was visible.
- The user could successfully access the folder.
- The **IT** folder was hidden because Access-Based Enumeration (ABE) prevented unauthorized folders from being displayed.



Step 8: Verify IT Department Access

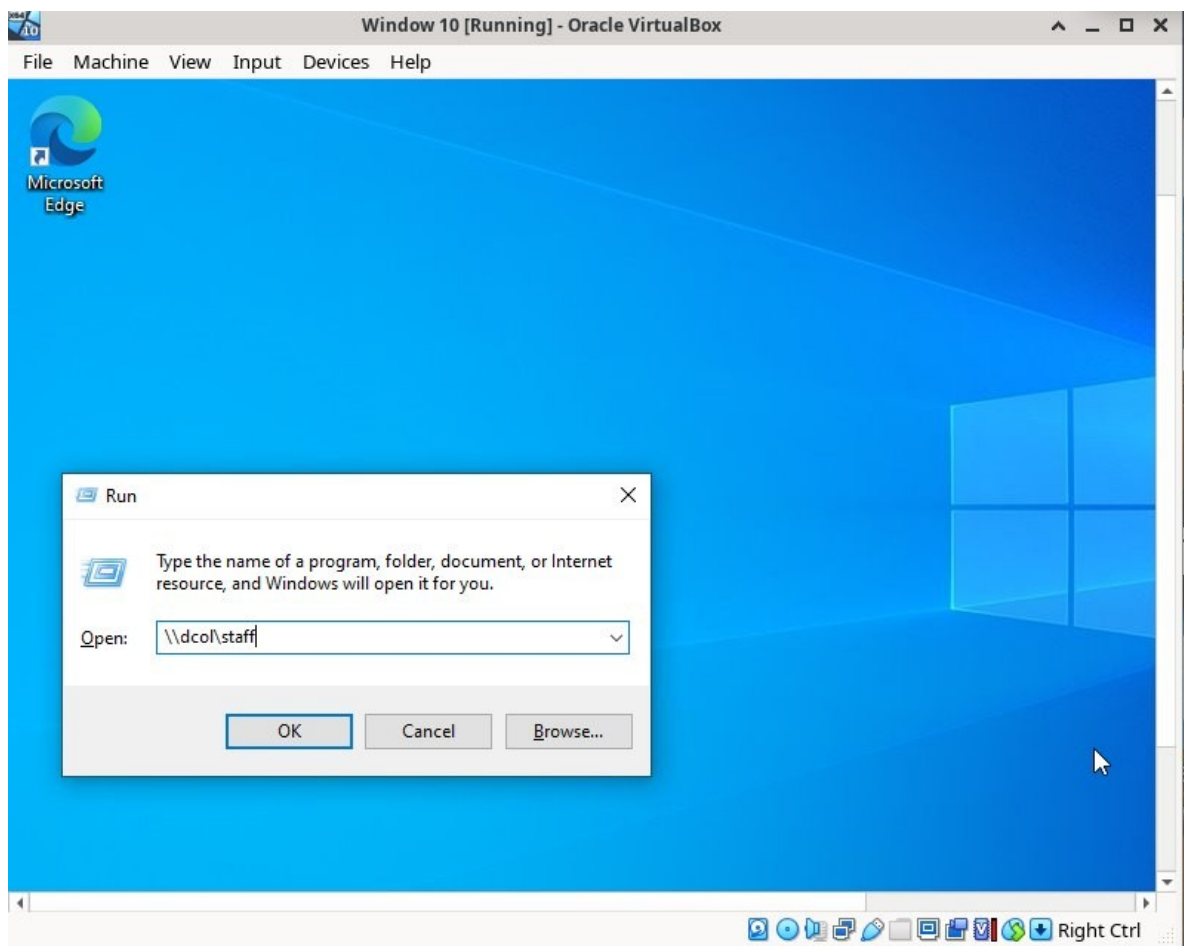
Logged into **CLIENT01** using the **sdavis** account.

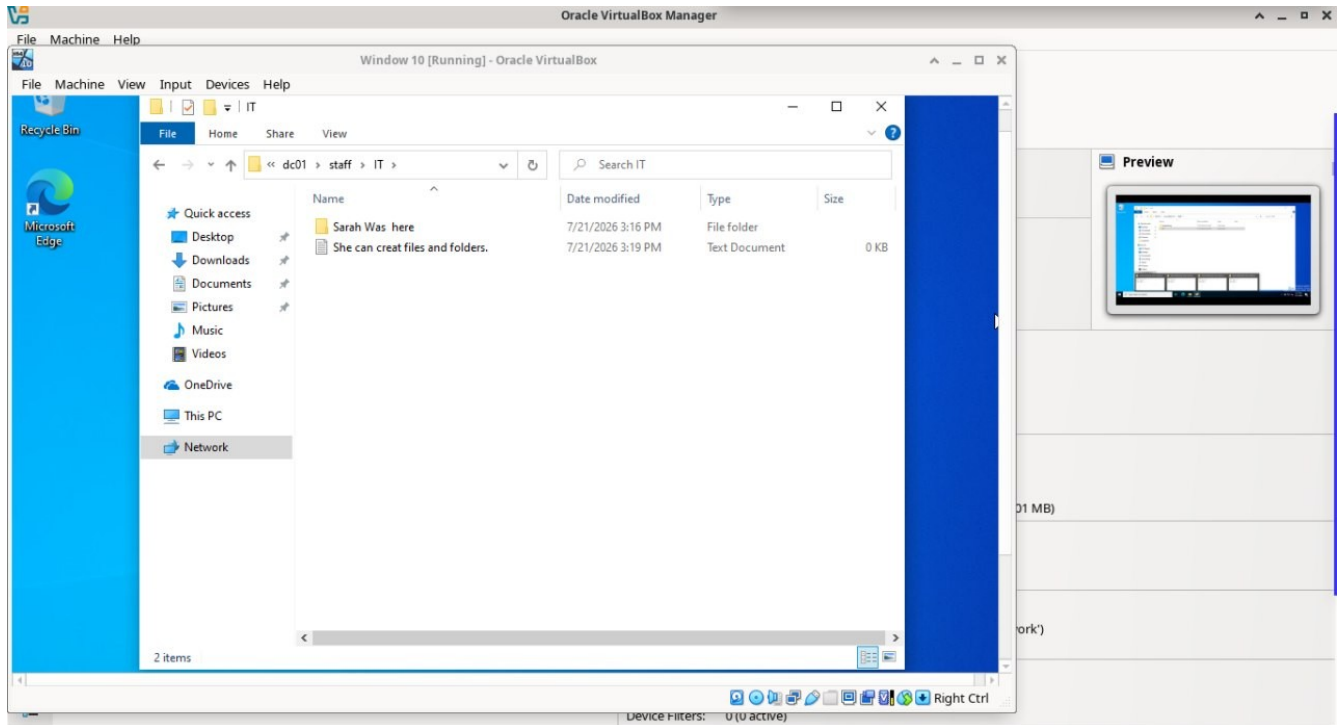
Opened:

\DC01\Staff

Verified that:

- The **IT** folder was visible.
- The user could successfully access the folder.
- The **Engineering** folder was hidden because of the configured NTFS permissions and Access-Based Enumeration.





Step 9: Verify Overall Security Controls

Reviewed the results of all authentication and file access tests to confirm that:

- Domain authentication was functioning correctly.
- Logon hour restrictions were enforced.
- Disabled accounts could not authenticate.
- Password change policies were enforced.
- Departmental file access matched the configured Active Directory security groups.
- Access-Based Enumeration successfully hid unauthorized folders.

User	Group	Can Login	Accessible Folder	Access Result (Reason)
jsmith	Engineering	Yes (Monday–Friday, 9:00 AM–5:00 PM only)	Engineering	Access Granted – Login occurred during the permitted logon hours, and the Engineering_Team group has Modify permission on the Engineering folder.
jsmith	Engineering	No (Outside permitted hours)	N/A	Access Denied – Active Directory blocked authentication because the configured logon hours allow access only from Monday to Friday, 9:00 AM–5:00 PM .
mjohnson	Engineering	No (Account Disabled)	N/A	Access Denied – The account was disabled in Active Directory and cannot authenticate until it is re-enabled by an administrator.
bwilliams	Engineering	Yes (After password change)	Engineering	Access Granted – After changing the password at first logon, the user authenticated successfully and accessed the Engineering folder as a member of Engineering_Team .
sdavis	IT	Yes	IT	Access Granted – The user is a member of IT_Team , which has Modify permission on the IT folder.
sdavis	IT	Yes (Attempted Engineering folder)	N/A	Access Denied – The Engineering folder has an Explicit Deny permission for IT_Team . Since Explicit Deny overrides Allow permissions, access was blocked.
tmiller	IT	Yes	IT	Access Granted – The user is a member of IT_Team and therefore has permission to access the IT folder.

Result

The client workstation successfully joined the **ot.local** domain and authenticated using Active Directory. Testing confirmed that logon hour restrictions, disabled account controls, password change enforcement, NTFS permissions, and Access-Based Enumeration all functioned correctly. Engineering users could access only Engineering resources, while IT users could access only IT resources, demonstrating effective implementation of Role-Based Access Control (RBAC).

Discussion

This exercise validated the complete Active Directory security configuration by testing authentication and authorization from a client workstation. Joining **CLIENT01** to the domain allowed centralized user authentication through the Domain Controller, while the configured security policies controlled when users could log in and which network resources they could access.

The successful implementation of Role-Based Access Control (RBAC) ensured that permissions were assigned according to departmental responsibilities rather than individual users. Access-Based Enumeration further strengthened security by hiding folders that users were not authorized to access, reducing unnecessary exposure of sensitive resources. These controls demonstrate how Active Directory can be used to enforce the Principle of Least Privilege in an Operational Technology environment.

Conclusion

The access control and security validation exercise successfully demonstrated that the Active Directory environment was operating as intended. Domain authentication, account management policies, NTFS permissions, and Access-Based Enumeration were all verified through practical testing on **CLIENT01**. The results confirmed that users could access only the resources appropriate to their assigned roles, providing a secure and well-managed Operational Technology domain environment.